MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGISTER, MULTIPORT, KEY CODE QQA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiport register module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1349 failures per million operating hours for class I or 0.2425 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QQA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/200-1 for class I and M28787/200-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

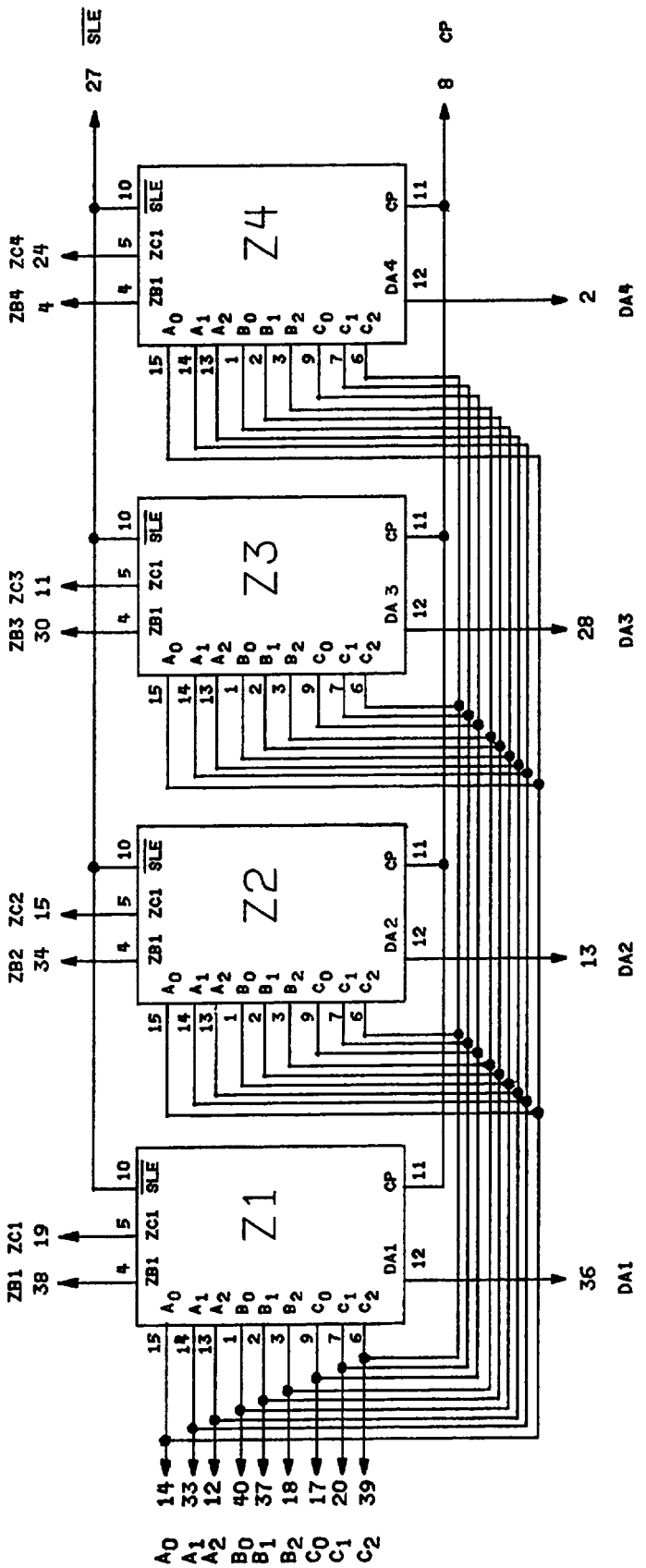
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTE:
Unused pins: 3, 5, 6, 7, 9, 16, 21, 22, 23, 25, 26, 29, 31, 32, 35.

FIGURE 1. QQA module function diagram, multiport register.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/ 0°C ±5°C		Class I 4/ 60°C ±5°C		Class II 4/ -55°C ±5°C		Class II 4/ 85°C ±5°C		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A17	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	III	S	← Must pass remaining tests →										
C1-C11	Input breakdown current	IV	S	0	4	0	4	0	4	0	4	0	4	mA
C12-C15	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D8	Output short circuit current	V	X	-10	-70	-10	-70	-10	-70	-10	-70	-10	-70	mA
E1-E11	Input current, high level	VI	X	0	108	0	108	0	108	0	108	0	108	µA
E12-E15	Input current, high level	VI	X	0	27	0	27	0	27	0	27	0	27	µA
F1-F11	Input current, low level	VII	X	0.0	-4.8	0.0	-4.8	0.0	-4.8	0.0	-4.8	0.0	-4.8	mA
F12-F15	Input current, low level	VII	X	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	mA
G1-G15	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1, H2	Power supply current, low level and high level	IX	X	0	540	0	540	0	540	0	540	0	540	mA
J1-J8	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K8	Output voltage, low level	XI	X	0	400	0	400	0	400	0	400	0	400	mV
L1-L122	Functional	XII		← See table XII →										
M1-M192	Propagation delay time (t_{PHL})	XIII	S	3	39	3	39	3	39	3	39	3	39	ns
N1-N192	Propagation delay time (t_{PLH})	XIV	S	4	44	4	44	4	44	4	44	4	44	ns
P1-P64	Propagation delay time (t_{PHL})	XV	S	5	54	5	54	5	54	5	54	5	54	ns
Q1-Q64	Propagation delay time (t_{PLH})	XVI	S	5	49	5	49	5	49	5	49	5	49	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R64	Propagation delay time (t_{PHL})	XVII	S	3	34	3	34	3	34	3	34	3	34	ns
S1-S64	Propagation delay time (t_{PLH})	XVIII	S	3	39	3	39	3	39	3	39	3	39	ns
T1-T256	Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL}, and t_{HH})	XIX	S	155	245	155	245	155	245	155	245	155	245	ns
U1-U128	Minimum clock pulse width (t_{CPL} and t_{CPH})	XX	S	205	295	205	295	205	295	205	295	205	295	ns
V	Power supply transient amplitude	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W8	Output crosstalk, high level	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
X1-X8	Output crosstalk, low level	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XXIII	S	1.98	2.42	1.98	2.42	1.98	2.42	1.98	2.42	1.98	2.42	µF
Z1-Z11	Input capacitance	XXIII	Q	8	60	8	60	8	60	8	60	8	60	pF
Z12-Z15	Input capacitance	XXIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Y shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

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	Class I	Class II	U
Maximum power dissipation (outputs not loaded)	2,178	2,178	m
Input current			
Low level (V_{IL} = 0.4 V)	-1.1 (DA1-DA4)	-1.1 (DA1-DA4)	m
	-4.4 (All others)	-4.4 (All others)	m
Input current			
High level (V_{IH} = 2.4 V)	27 (DA1-DA4)	27 (DA1-DA4)	µ
	108 (All others)	108 (All others)	µ
Output drive current			
Low level (V_{OL} = 0.4 V)	16	16	m
High level (V_{OH} = 2.4 V)	-0.8	-0.8	m
Setup and hold time			
t_{SET} (high data to clock)	20	20	n
t_{HOLD} (high data to clock)	0.0	0.0	n
t_{SET} (low data to clock)	12	12	n
t_{HOLD} (low data to clock)	-8	-8	n
t_{SET} (A address to clock)	10	10	n
t_{HOLD} (A address to clock)	0.0	0.0	n

6.5 Input to output relationship. Each QQA module has the input to output relationship of figure 2 for positive logic.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

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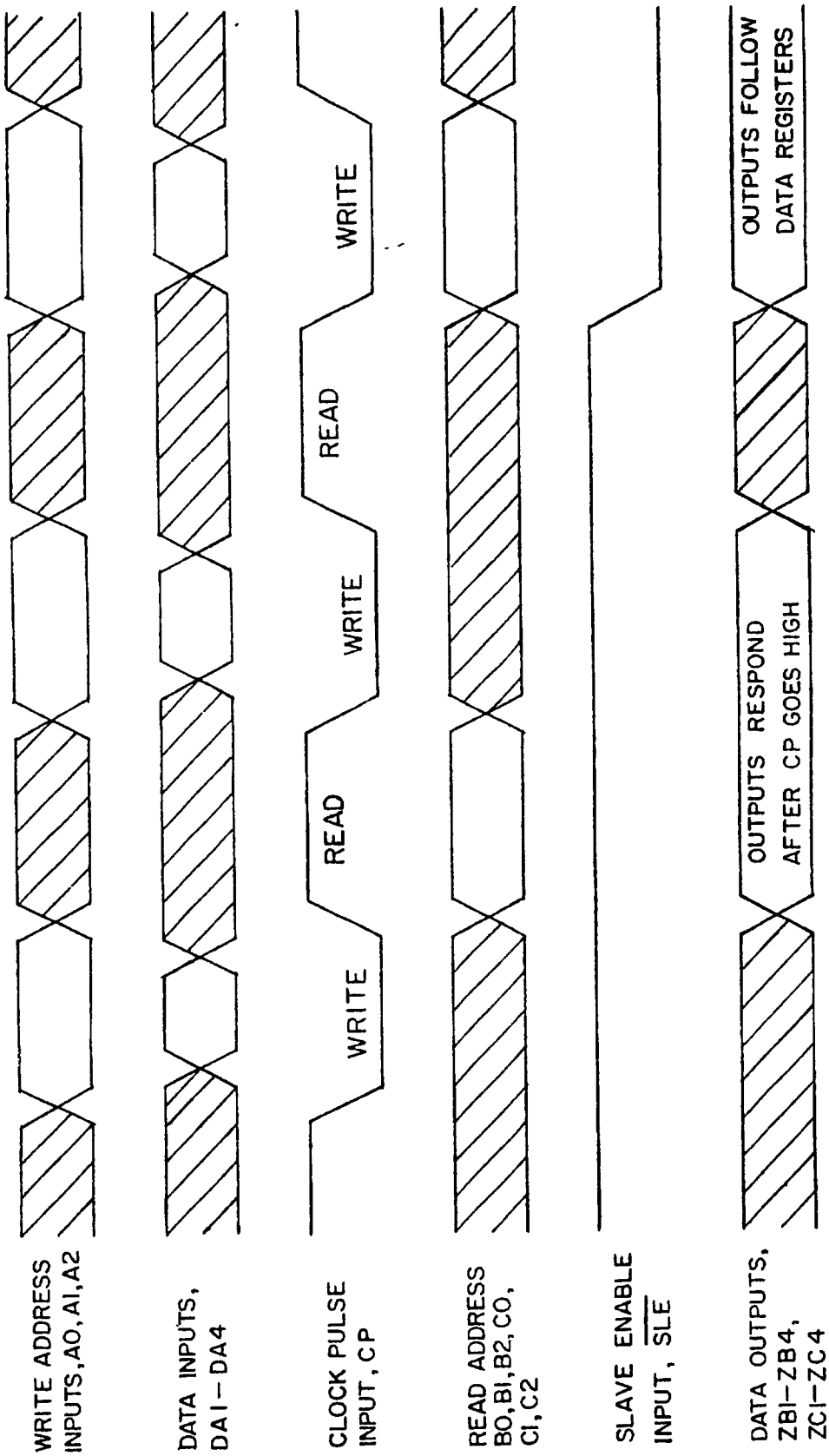


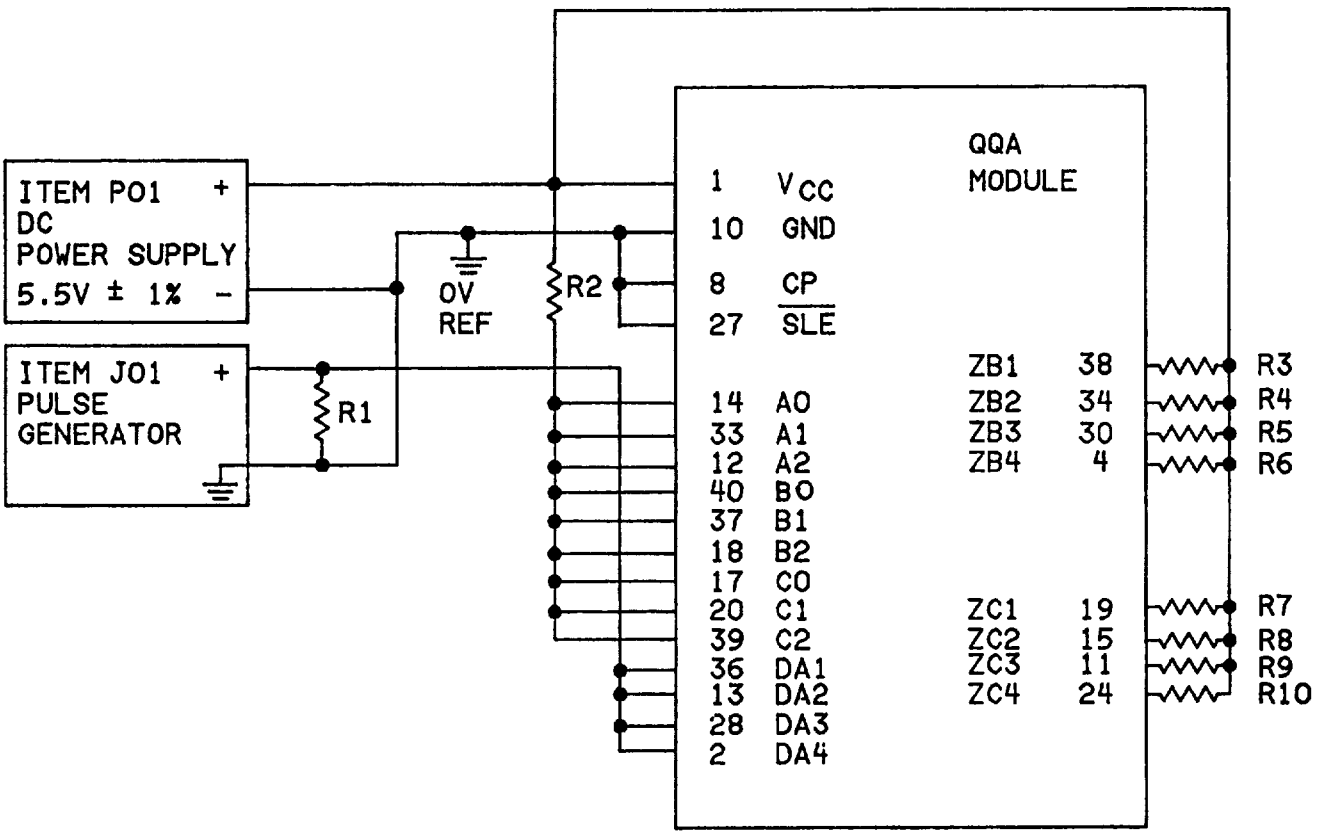
FIGURE 2. QQA module input to output relationship.

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NOTES:
1. The waveforms shown represent input-output relationships and do not depict timing requirements.
2. The QQA module is configured as eight working registers of four bits each. Data may be written into any of eight locations and read from any two of the eight registers simultaneously. While the clock pulse (CP) is low the outputs are held steady, but the data inputs (DA1-DA4) are permitted to enter the register selected by the write adders (A0-A2) inputs. The next CP transition from low to high latches the registers in the present states. This rising CP edge also connects each of the outputs (ZB1-ZB4 and ZC1-ZC4) to the register selected by the read adders (B0-B2 and C0-C2) lines causing the register contents to be reflected on the outputs. If the slave enable ($\overline{SLE}$) input is held low the outputs are continuously enabled, allowing immediate transfer of selected register contents to the outputs.

FIGURE 2. QQA module input to output relationship - Continued.

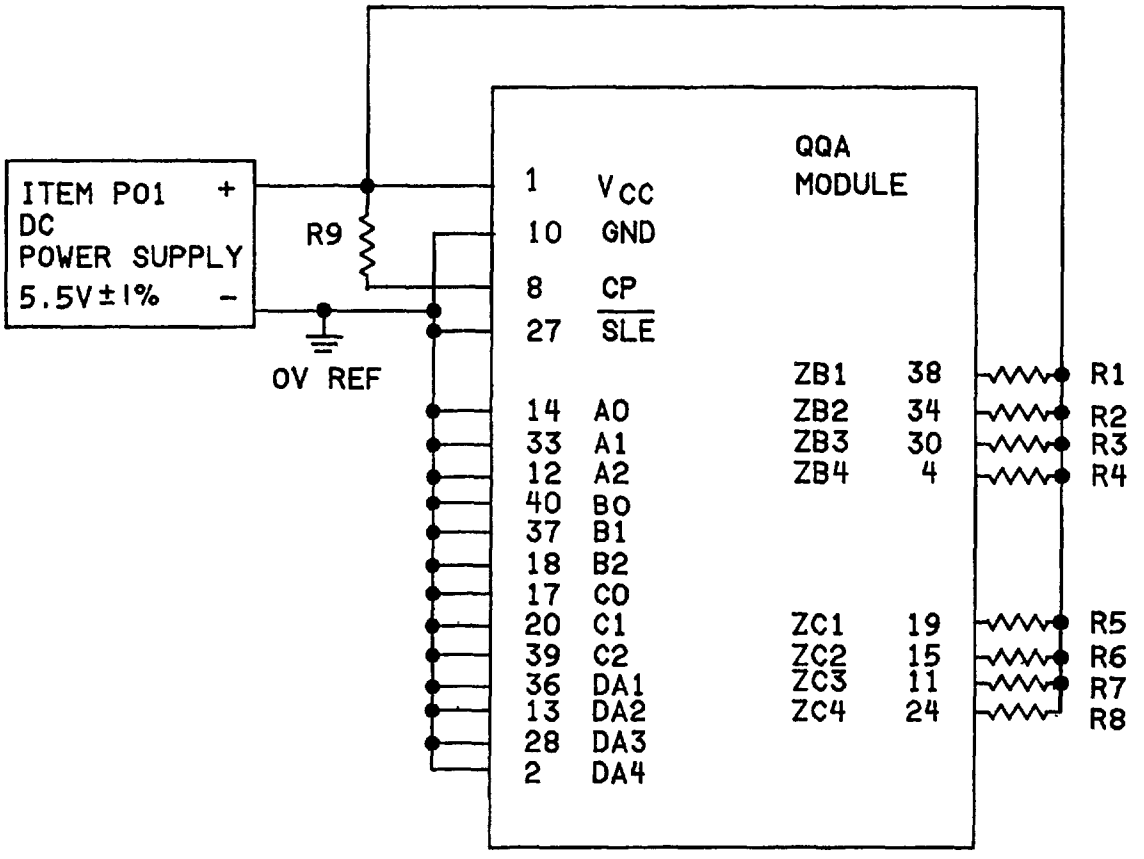
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NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. R2 = 619 ±1%, 1/4 W.
3. R3 - R10 = 316Ω ±1%, 1/4 W.
4. Item J01 setup:
   PW = 250 ±25 ns.
   PRP = 500 ±50 ns.
   PA = 3.0 ±0.1 V.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 3. Operating temperature and life test circuit.

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NOTE: R1 - R8 = 316Ω ±1%, 1/4 W.
R9 = 1 kΩ ±1%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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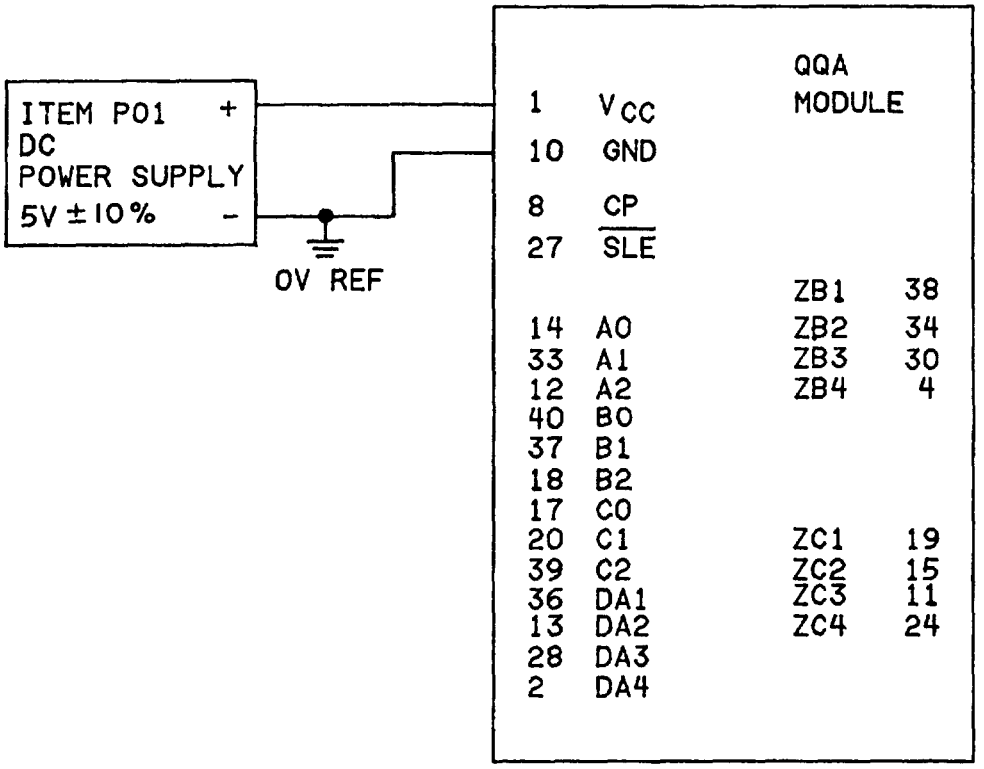


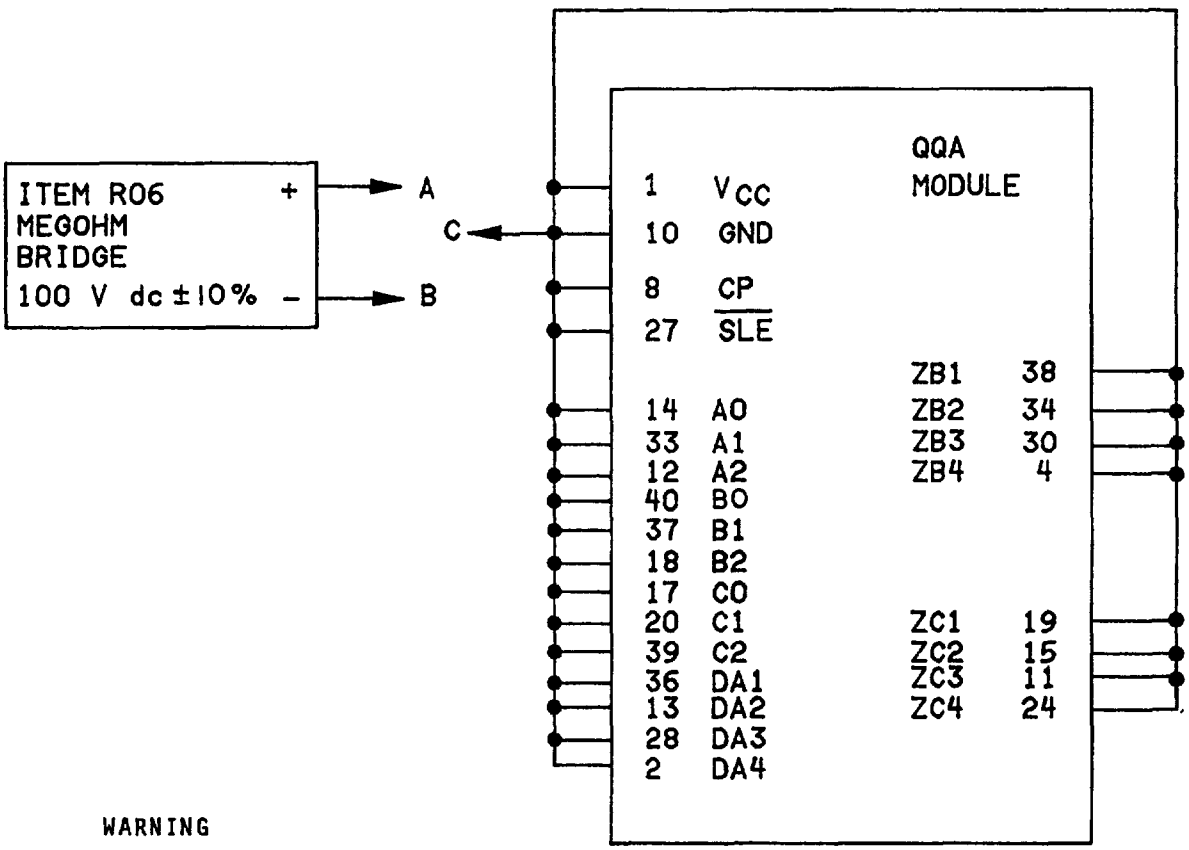
FIGURE 5. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C,3,5,6,7,9,16,21,22,23,25,26,29,31,32,35	Frame	10	∞	MΩ	6
A2	Isolation, used pins to unused pins	C	3,5,6,7,9,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A3	Isolation, unused pins to unused pins	3	5,6,7,9,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A4	"	5	3,6,7,9,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A5	"	6	3,5,7,9,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A6	"	7	3,5,6,9,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A7	"	9	3,5,6,7,16,21,22,23,25,26,29,31,32,35	"	"	"	"
A8	"	16	3,5,6,7,9,21,22,23,25,26,29,31,32,35	"	"	"	"
A9	"	21	3,5,6,7,9,16,22,23,25,26,29,31,32,35	"	"	"	"
A10	"	22	3,5,6,7,9,16,21,23,25,26,29,31,32,35	"	"	"	"
A11	"	23	3,5,6,7,9,16,21,22,25,26,29,31,32,35	"	"	"	"
A12	"	25	3,5,6,7,9,16,21,22,23,26,29,31,32,35	"	"	"	"
A13	"	26	3,5,6,7,9,16,21,22,23,25,29,31,32,35	"	"	"	"
A14	"	29	3,5,6,7,9,16,21,22,23,25,26,31,32,35	"	"	"	"
A15	"	31	3,5,6,7,9,16,21,22,23,25,26,29,32,35	"	"	"	"
A16	"	32	3,5,6,7,9,16,21,22,23,25,26,29,31,35	"	"	"	"
A17	"	35	3,5,6,7,9,16,21,22,23,25,26,29,31,32	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

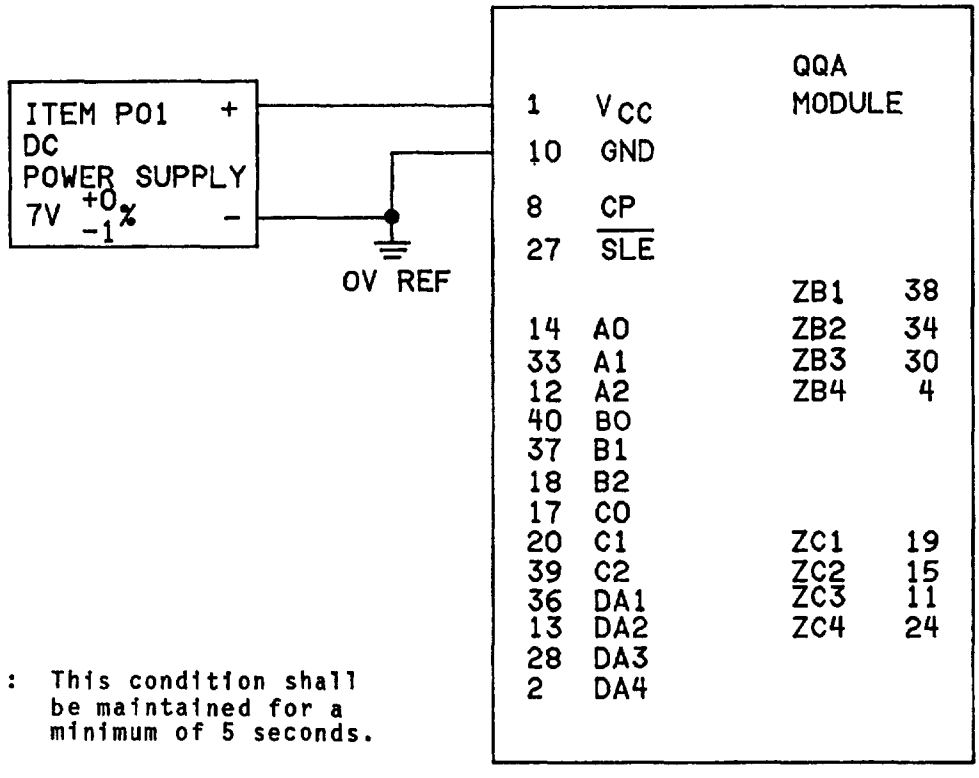
FIGURE 6. Isolation test circuit for table II.

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TABLE III. V_{CC} stress test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
B	V_{CC} stress	← Must pass remaining tests →				7

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NOTE: This condition shall be maintained for a minimum of 5 seconds.

FIGURE 7. V_{CC} stress test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	CP	Input breakdown current	8	27,14,33,12,40,37,18,17, 20,39,36,13,28,2	0	4	mA	8
C2	$\overline{SLE}$	"	27	8,14,33,12,40,37,18,17, 20,39,36,13,28,2	"	"	"	"
C3	A0	"	14	8,27,33,12,40,37,18,17, 20,39,36,13,28,2	"	"	"	"
C4	A1	"	33	8,27,14,12,40,37,18,17, 20,39,36,13,28,2	"	"	"	"
C5	A2	"	12	8,27,14,33,40,37,18,17, 20,39,36,13,28,2	"	"	"	"
C6	B0	"	40	8,27,14,33,12,37,18,17, 20,39,36,13,28,2	"	"	"	"
C7	B1	"	37	8,27,14,33,12,40,18,17, 20,39,36,13,28,2	"	"	"	"
C8	B2	"	18	8,27,14,33,12,40,37,17, 20,39,36,13,28,2	"	"	"	"
C9	C0	"	17	8,27,14,33,12,40,37,18, 20,39,36,13,28,2	"	"	"	"
C10	C1	"	20	8,27,14,33,12,40,37,18, 17,39,36,13,28,2	"	"	"	"
C11	C2	"	39	8,27,14,33,12,40,37,18, 17,20,36,13,28,2	"	"	"	"
C12	DA1	"	36	8,27,14,33,12,40,37,18, 17,20,39,13,28,2	"	1	"	"
C13	DA2	"	13	8,27,14,33,12,40,37,18, 17,20,39,36,28,2	"	"	"	"
C14	DA3	"	28	8,27,14,33,12,40,37,18, 17,20,39,36,13,2	"	"	"	"
C15	DA4	"	2	8,27,14,33,12,40,37,18, 17,20,39,36,13,28	"	"	"	"

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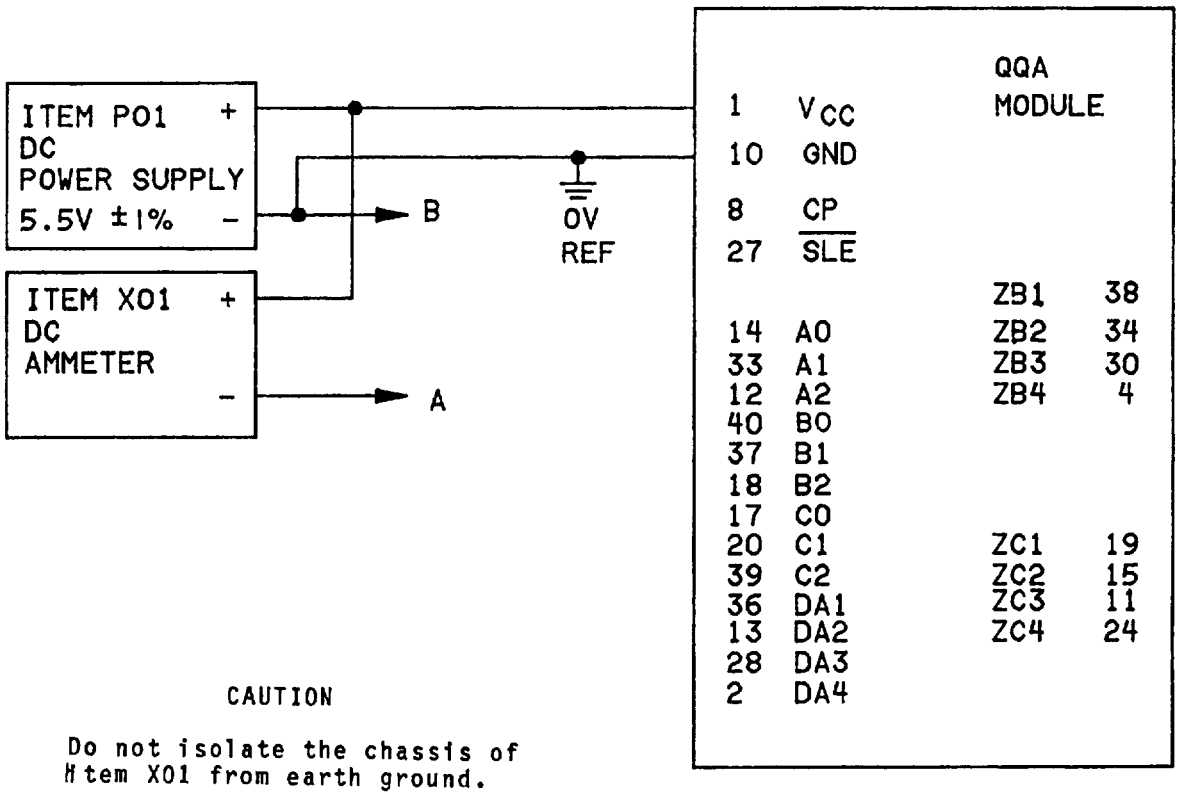


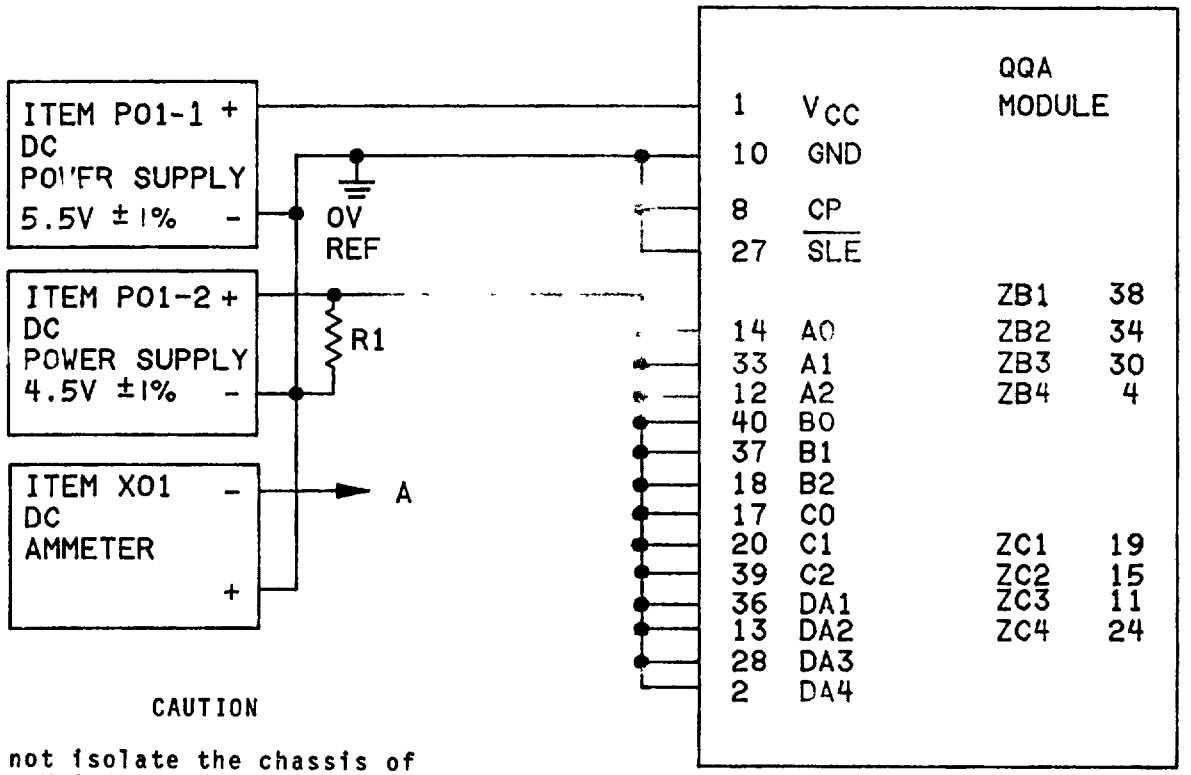
FIGURE 8. Input breakdown current test circuit for table IV.

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TABLE V. Output short circuit current test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	ZB1	Output short circuit current	38	-10	-70	mA	9
D2	ZB2	"	34	"	"	"	"
D3	ZB3	"	30	"	"	"	"
D4	ZB4	"	4	"	"	"	"
D5	ZC1	"	19	"	"	"	"
D6	ZC2	"	15	"	"	"	"
D7	ZC3	"	11	"	"	"	"
D8	ZC4	"	24	"	"	"	"

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

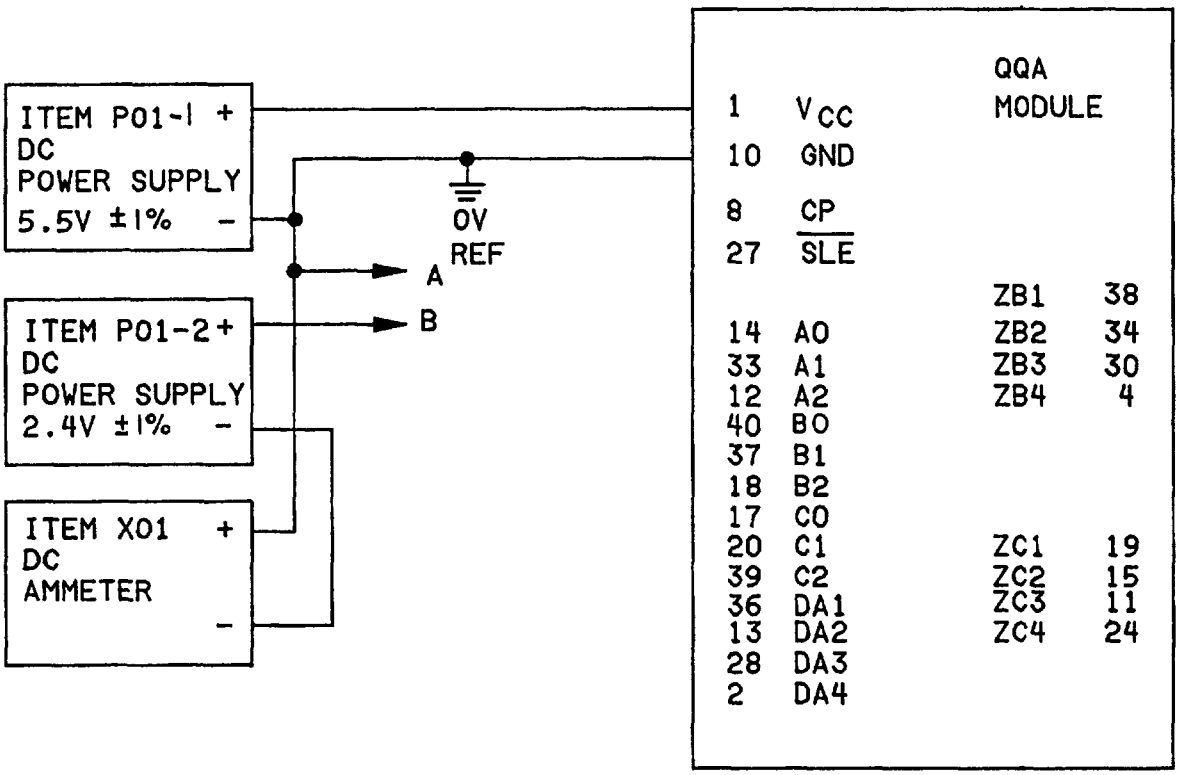
FIGURE 9. Output short circuit current test circuit for table V.

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TABLE VI. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	CP	Input current, high level	27,14,33,12,40,37,18,17, 20,39,36,13,28,2	8	0	108	µA	10
E2	$\overline{SLE}$	"	8,14,33,12,40,37,18,17, 20,39,36,13,28,2	27	"	"	"	"
E3	A0	"	8,27,33,12,40,37,18,17, 20,39,36,13,28,2	14	"	"	"	"
E4	A1	"	8,27,14,12,40,37,18,17, 20,39,36,13,28,2	33	"	"	"	"
E5	A2	"	8,27,14,33,40,37,18,17, 20,39,36,13,28,2	12	"	"	"	"
E6	B0	"	8,27,14,33,12,37,18,17, 20,39,36,13,28,2	40	"	"	"	"
E7	B1	"	8,27,14,33,12,40,18,17, 20,39,36,13,28,2	37	"	"	"	"
E8	B2	"	8,27,14,33,12,40,37,17, 20,39,36,13,28,2	18	"	"	"	"
E9	C0	"	8,27,14,33,12,40,37,18, 20,39,36,13,28,2	17	"	"	"	"
E10	C1	"	8,27,14,33,12,40,37,18, 17,39,36,13,28,2	20	"	"	"	"
E11	C2	"	8,27,14,33,12,40,37,18, 17,20,36,13,28,2	39	"	"	"	"
E12	DA1	"	8,27,14,33,12,40,37,18, 17,20,39,13,28,2	36	"	27	"	"
E13	DA2	"	8,27,14,33,12,40,37,18, 17,20,39,36,28,2	13	"	"	"	"
E14	DA3	"	8,27,14,33,12,40,37,18, 17,20,39,36,13,2	28	"	"	"	"
E15	DA4	"	8,27,14,33,12,40,37,18, 17,20,39,36,13,28	2	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

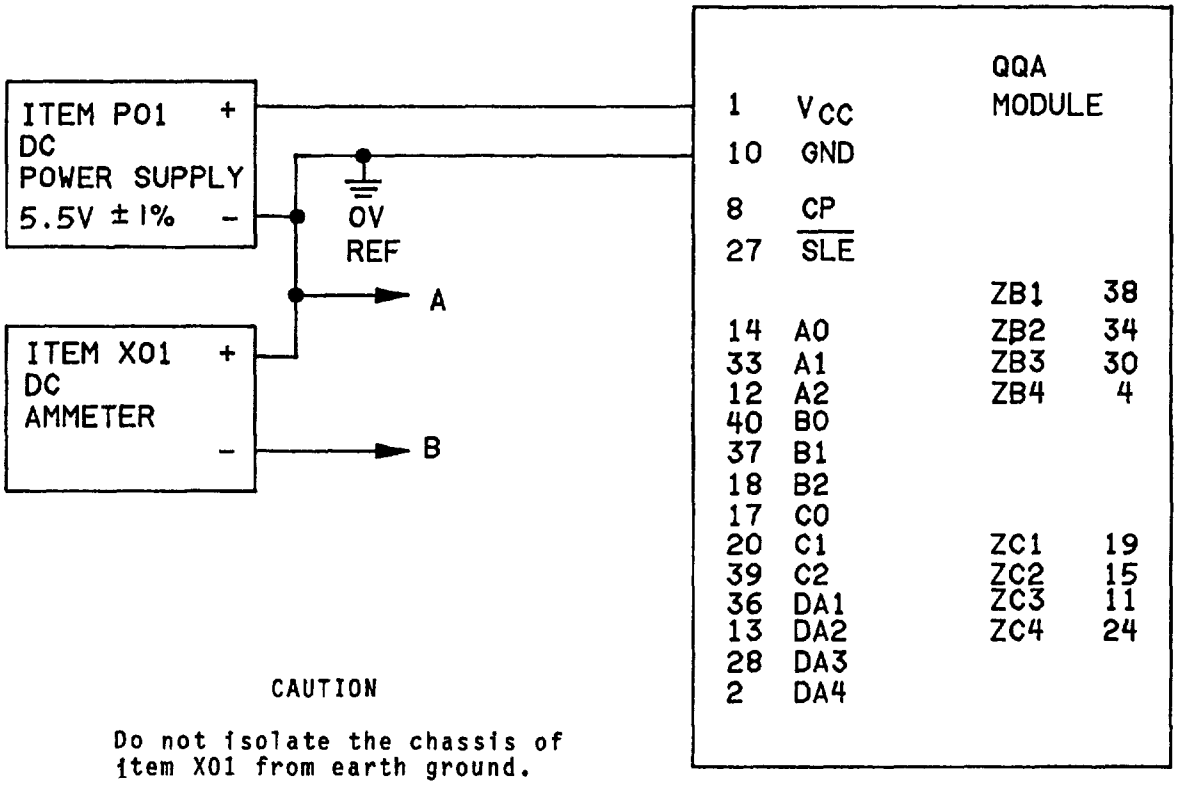
FIGURE 10. Input current (high-level) test circuit for table VI.

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TABLE VII. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	CP	Input current, low level	27,14,33,12,40,37,18,17, 20,39,36,13,28,2	8	0.0	-4.8	mA	11
F2	$\overline{SLE}$	"	8,14,33,12,40,37,18,17, 20,39,36,13,28,2	27	"	"	"	"
F3	A0	"	8,27,33,12,40,37,18,17, 20,39,36,13,28,2	14	"	"	"	"
F4	A1	"	8,27,14,12,40,37,18,17, 20,39,36,13,28,2	33	"	"	"	"
F5	A2	"	8,27,14,33,40,37,18,17, 20,39,36,13,28,2	12	"	"	"	"
F6	B0	"	8,27,14,33,12,37,18,17, 20,39,36,13,28,2	40	"	"	"	"
F7	B1	"	8,27,14,33,12,40,18,17, 20,39,36,13,28,2	37	"	"	"	"
F8	B2	"	8,27,14,33,12,40,37,17, 20,39,36,13,28,2	18	"	"	"	"
F9	C0	"	8,27,14,33,12,40,37,18, 20,39,36,13,28,2	17	"	"	"	"
F10	C1	"	8,27,14,33,12,40,37,18, 17,39,36,13,28,2	20	"	"	"	"
F11	C2	"	8,27,14,33,12,40,37,18, 17,20,36,13,28,2	39	"	"	"	"
F12	DA1	"	8,27,14,33,12,40,37,18, 17,20,39,13,28,2	36	"	-1.2	"	"
F13	DA2	"	8,27,14,33,12,40,37,18, 17,20,39,36,28,2	13	"	"	"	"
F14	DA3	"	8,27,14,33,12,40,37,18, 17,20,39,36,13,2	28	"	"	"	"
F15	DA4	"	8,27,14,33,12,40,37,18, 17,20,39,36,13,28	2	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

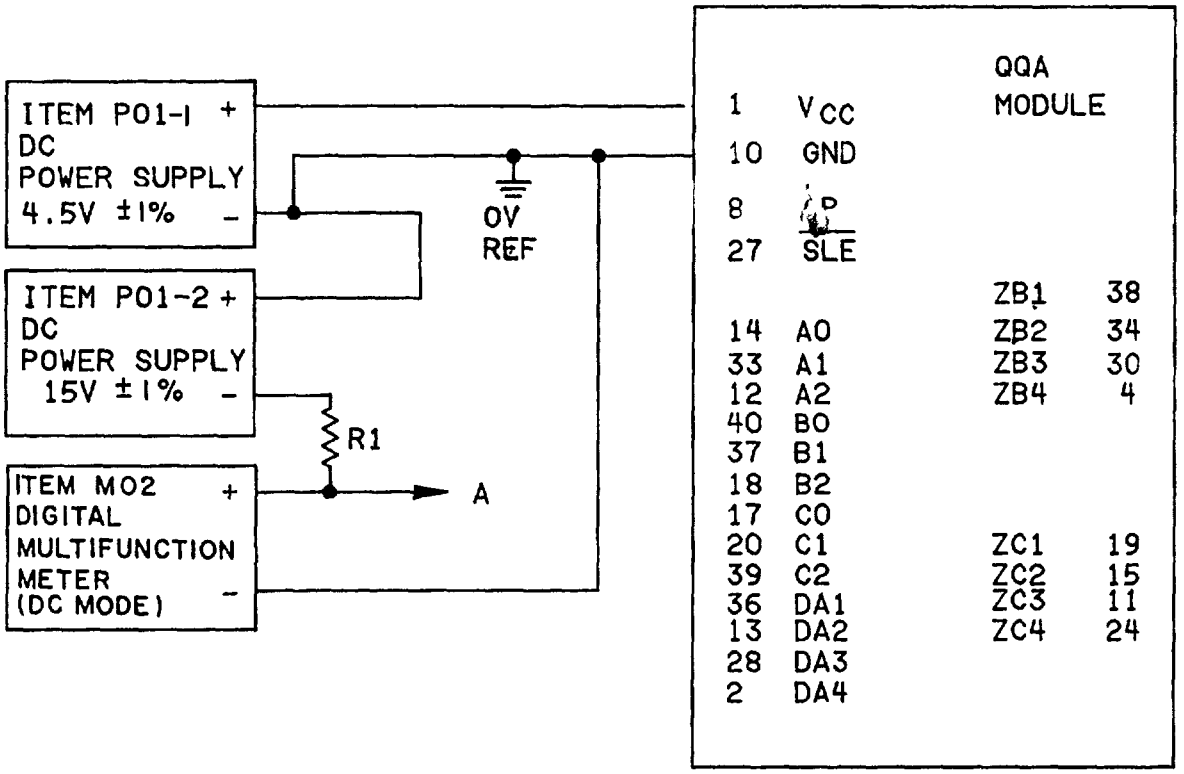
FIGURE 11. Input current (low-level) test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
G1	CP	Input clamp diode voltage	8	-0.4	-1.5	V	12
G2	$\overline{SLE}$	"	27	"	"	"	"
G3	A0	"	14	"	"	"	"
G4	A1	"	33	"	"	"	"
G5	A2	"	12	"	"	"	"
G6	B0	"	40	"	"	"	"
G7	B1	"	37	"	"	"	"
G8	B2	"	18	"	"	"	"
G9	C0	"	17	"	"	"	"
G10	C1	"	20	"	"	"	"
G11	C2	"	39	"	"	"	"
G12	DA1	"	36	"	"	"	"
G13	DA2	"	13	"	"	"	"
G14	DA3	"	28	"	"	"	"
G15	DA4	"	2	"	"	"	"

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NOTE: R1 = 1130Ω ±1%, 1/4 W.

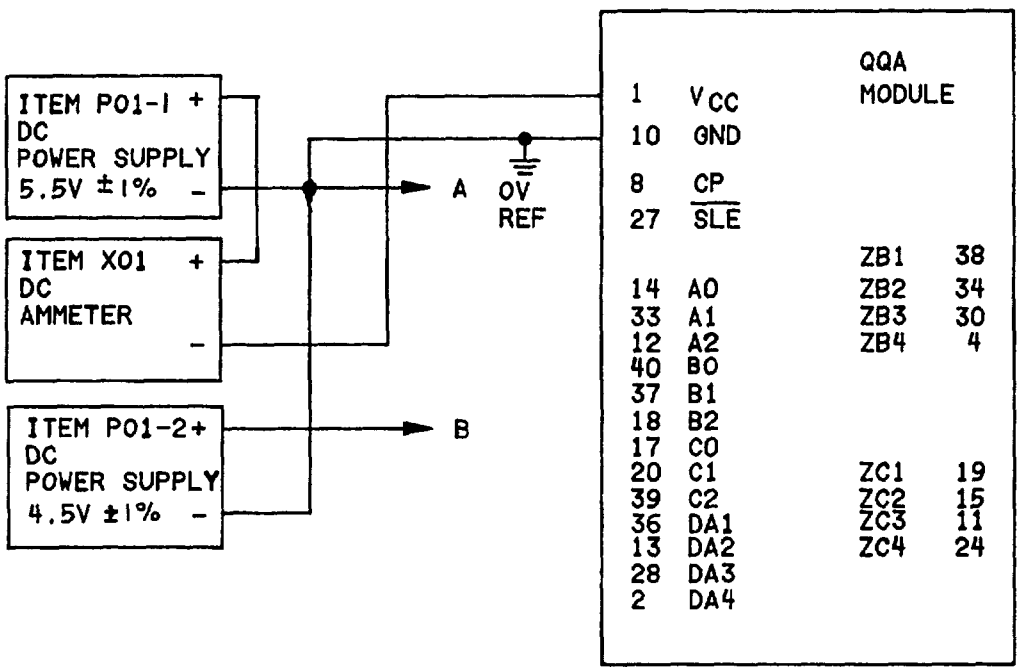
FIGURE 12. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Power supply current, low level	8,27,14,33,12, 40,37,18,17,20, 39,36,13,28,2	Open	0	540	mA	13
H2	Power supply current, high level	8,27	14,33,12,40,37, 18,17,20,39,36, 13,28,2	0	540	mA	13

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table IX.

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TABLE X. Output voltage (high-level) test.

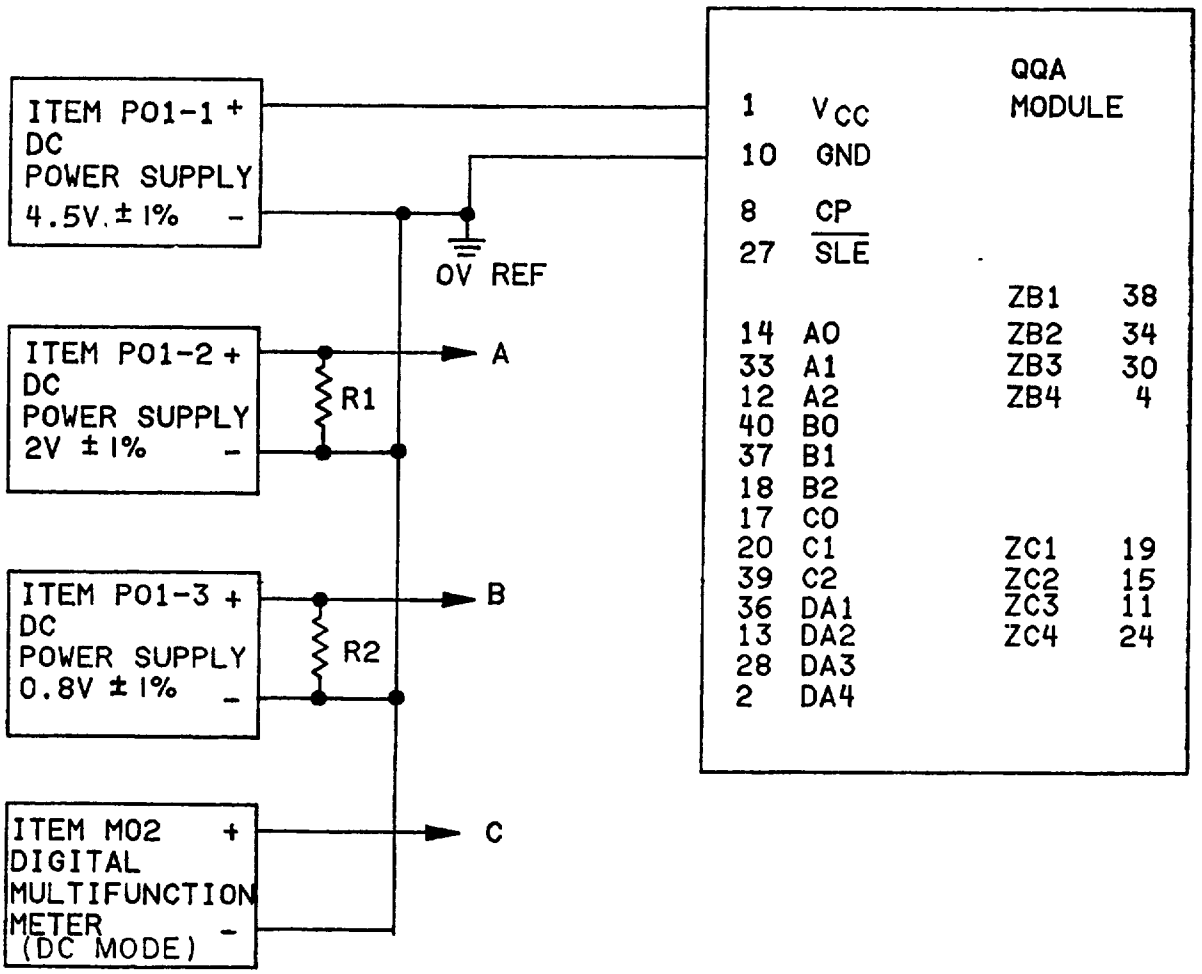
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	ZB1 Output voltage, high level	36	8,27,14,33,12,40, 37,18,17,20,39, 13,28,2	38	2.4	4.5	V	14
J2	ZB2 "	13,14,40,17	8,27,33,12,37,18, 20,39,36,28,2	34	"	"	"	"
J3	ZB3 "	28,33,37,20	8,27,14,12,40,18, 17,39,36,13,2	30	"	"	"	"
J4	ZB4 "	2,14,33,40,37, 17,20	8,27,12,18,39,36, 13,28	4	"	"	"	"
J5	ZC1 "	36,12,18,39	8,27,14,33,40,37, 17,20,13,28,2	19	"	"	"	"
J6	ZC2 "	13,12,18,39,14, 40,17	8,27,12,37,20, 36,28,2	15	"	"	"	"
J7	ZC3 "	28,33,12,37,18, 20,39	8,27,14,40,17, 36,13,2	11	"	"	"	"
J8	ZC4 "	2,14,33,12,40, 37,18,17,20,39	8,27,36,13,28	24	"	"	"	"

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TABLE XI. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	ZB1	Output voltage, low level	Open	8,27,14,33,12,40,37,18,17,20,39,13,28,2,36	38	0	400	mV	14
K2	ZB2	"	14,40,17	8,27,33,12,37,18,20,39,36,28,2,13	34	"	"	"	"
K3	ZB3	"	33,37,20	8,27,14,12,40,18,17,39,36,13,2,28	30	"	"	"	"
K4	ZB4	"	14,33,40,37,17,20	8,27,12,18,39,36,13,28,2	4	"	"	"	"
K5	ZC1	"	12,18,39	8,27,14,33,40,37,17,20,13,28,2,36	19	"	"	"	"
K6	ZC2	"	12,18,39,14,40,17	8,27,12,37,20,36,28,2,13	15	"	"	"	"
K7	ZC3	"	33,12,37,18,20,39	8,27,14,40,17,36,13,2,28	11	"	"	"	"
K8	ZC4	"	14,33,12,40,37,18,17,20,39	8,27,36,13,28,2	24	"	"	"	"

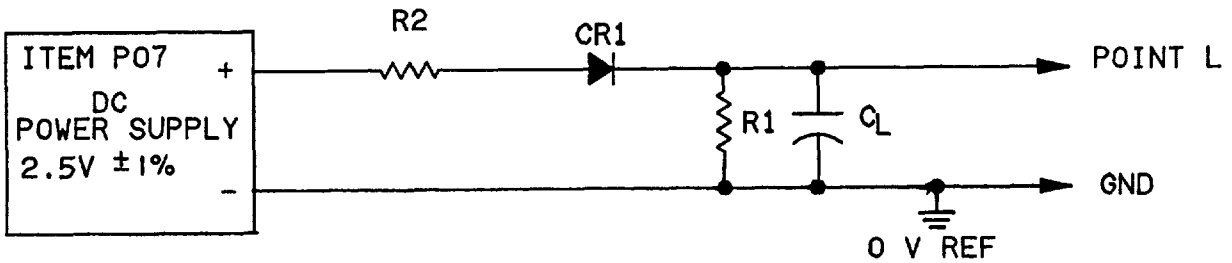
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. Load all outputs with load as shown on figure 15.

FIGURE 14. Output voltage (high-level and low-level) test circuit for tables X and XI.

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NOTES:
1. R1 = 5110Ω ±1%, 1/4 W.
2. R2 = Select value for current flow of 16 mA ±5% out of point L, with point L held at 0.4 V ±1%.
3. CR1 = 1N4150 or equivalent.
4. C_L = 50 pF ±5%.
5. C_L is not required for dc electrical tests.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 15. QQA logic load circuit.

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TABLE XII. Functional test.

Test condition	Parameter	Connect															Outputs								Test figure
		Pin 14 to	Pin 33 to	Pin 12 to	Pin 40 to	Pin 37 to	Pin 18 to	Pin 17 to	Pin 20 to	Pin 39 to	Pin 36 to	Pin 13 to	Pin 28 to	Pin 2 to	Pin 8 to	Pin 27 to	Pin 38	Pin 34	Pin 30	Pin 4	Pin 19	Pin 15	Pin 11	Pin 24	
L1	Functional	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	L	L	L	L	L	L	L	L	16
L2	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L3	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L4	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L5	"	B	B	A	B	B	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L6	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L7	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L8	"	A	"	"	A	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L9	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"
L10	"	"	"	"	"	"	"	"	"	"	A	A	A	A	"	"	"	"	"	"	"	"	"	"	"
L11	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"
L12	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L13	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"
L14	"	B	"	"	B	"	"	B	"	"	"	"	"	"	"	"	L	L	L	L	L	L	L	L	"
L15	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L16	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L17	"	A	A	B	A	A	B	A	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L18	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L19	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L20	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L21	"	A	A	A	A	A	A	A	A	A	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L22	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"
L23	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"									"
L24	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"									"

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TABLE XII. Functional test - Continued.

Test condition	Parameter	Connect															Outputs								Test figure
		Pin 14 to	Pin 33 to	Pin 12 to	Pin 40 to	Pin 37 to	Pin 18 to	Pin 17 to	Pin 20 to	Pin 39 to	Pin 36 to	Pin 13 to	Pin 28 to	Pin 2 to	Pin 8 to	Pin 27 to	Pin 38	Pin 34	Pin 30	Pin 4	Pin 19	Pin 15	Pin 11	Pin 24	
L25	Functional	B	A	A	B	A	A	B	A	A	A	A	A	A	A	A									16
L26	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"									"
L27	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"									"
L28	"	A	"	"	A	"	"	A	"	"	"	"	"	"	"	"									"
L29	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"									"
L30	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"									"
L31	"	B	A	B	B	A	B	B	A	B	"	"	"	"	"	"									"
L32	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"									"
L33	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"									"
L34	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"									"
L35	"	B	A	A	B	A	A	B	A	A	"	"	"	"	"	"									"
L36	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"									"
L37	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"									"
L38	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"									"
L39	"	A	B	"	A	B	"	A	B	"	A	A	A	A	A	"									"
L40	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"									"
L41	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"									"
L42	"	B	"	"	B	"	"	B	"	"	"	"	"	"	B	"									"
L43	"	A	A	A	A	A	A	A	A	A	A	A	A	A	A	"	L	L	L	L	L	L	L	L	"
L44	"	B	A	A	B	A	A	B	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L45	"	A	B	B	A	B	B	A	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L46	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L47	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L48	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

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TABLE XII. Functional test - Continued.

Test condition	Parameter	Connect															Outputs								Test figure
		Pin 14 to	Pin 33 to	Pin 12 to	Pin 40 to	Pin 37 to	Pin 18 to	Pin 17 to	Pin 20 to	Pin 39 to	Pin 36 to	Pin 13 to	Pin 28 to	Pin 2 to	Pin 8 to	Pin 27 to	Pin 38	Pin 34	Pin 30	Pin 4	Pin 19	Pin 15	Pin 11	Pin 24	
L49	Functional	A	B	A	A	B	A	A	B	A	A	A	A	A	A	A	H	H	H	H	H	H	H	H	16
L50	"	"	"	"	"	"	"	"	"	"	B	B	B	B	A	"	"	"	"	"	"	"	"	"	"
L51	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L52	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L53	"	B	"	"	B	"	"	B	"	"	A	A	A	A	A	"	"	"	"	"	"	"	"	"	"
L54	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L55	"	B	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"
L56	"	A	"	"	A	"	"	A	"	"	"	"	"	"	"	"	L	L	L	L	L	L	L	L	"
L57	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L58	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L59	"	B	B	B	B	B	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L60	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L61	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L62	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L63	"	B	B	A	B	B	A	B	B	A	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L64	"	"	"	"	"	"	"	"	"	"	B	B	B	B	A	"	"	"	"	"	"	"	"	"	"
L65	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L66	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L67	"	A	A	B	A	A	B	A	A	B	A	A	A	A	A	"	"	"	"	"	"	"	"	"	"
L68	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L69	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"
L70	"	B	"	"	B	"	"	B	"	"	"	"	"	"	"	"	L	L	L	L	L	L	L	L	"
L71	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L72	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

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TABLE XII. Functional test - Continued.

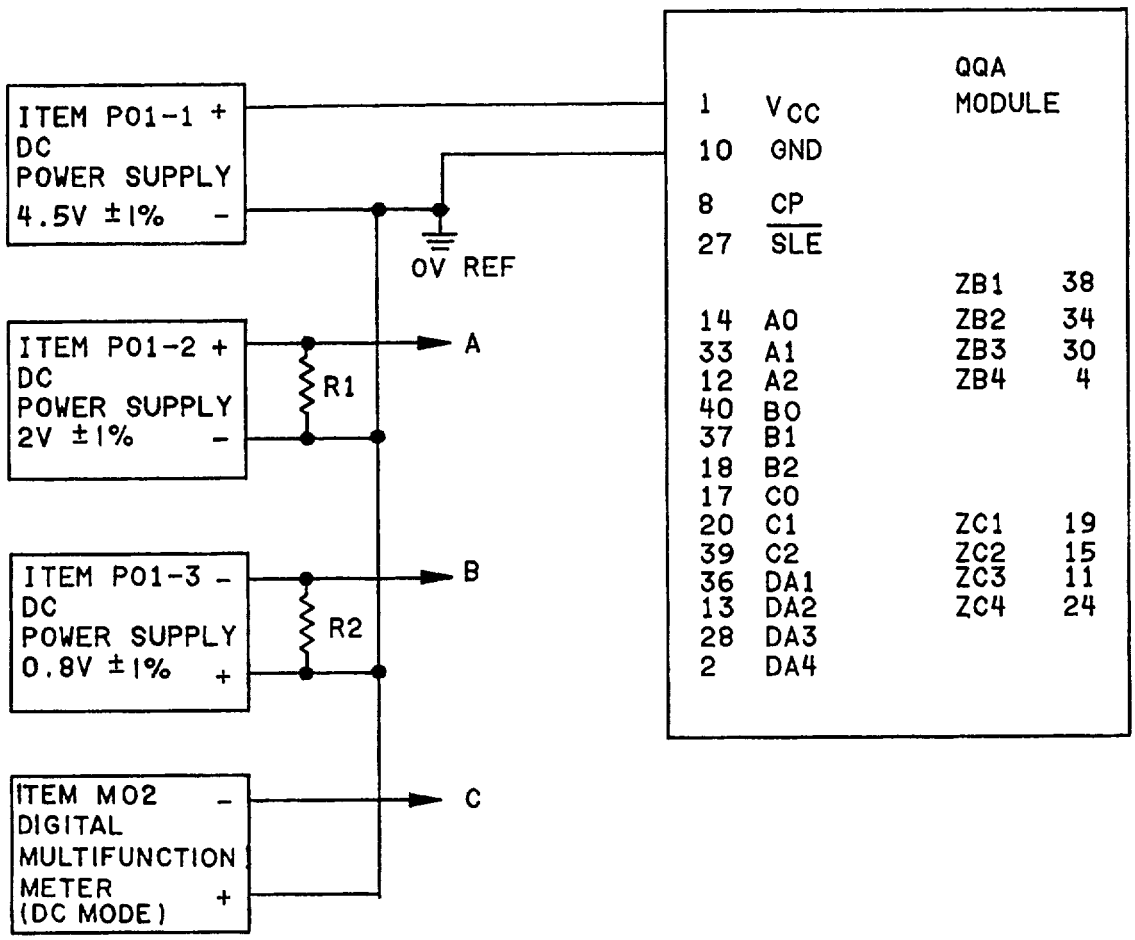
Test condition	Parameter	Connect															Outputs								Test figure
		Pin 14 to	Pin 33 to	Pin 12 to	Pin 40 to	Pin 37 to	Pin 18 to	Pin 17 to	Pin 20 to	Pin 39 to	Pin 36 to	Pin 13 to	Pin 28 to	Pin 2 to	Pin 8 to	Pin 27 to	Pin 38	Pin 34	Pin 30	Pin 4	Pin 19	Pin 15	Pin 11	Pin 24	
L73	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	L	L	L	L	L	L	L	16
L74	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L75	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L76	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"		"	"	"	"	"	"	"	"
L77	"	A	A	B	A	A	B	A	A	B	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L78	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"
L79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L81	"	B	"	"	B	"	"	B	"	"	A	A	A	A	A	"	"	"	"	"	"	"	"	"	"
L82	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L83	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"
L84	"	A	"	"	A	"	"	A	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L85	"	B	B	B	B	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L86	"	A	B	B	A	B	B	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L87	"	B	A	A	B	A	A	B	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L88	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L89	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L90	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L91	"	B	A	B	B	A	B	B	A	B	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L92	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"
L93	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L94	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L95	"	A	B	"	A	B	"	A	B	"	A	A	A	A	A	"	"	"	"	"	"	"	"	"	"
L96	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L97	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"

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TABLE XII. Functional test - Continued.

Test condition	Parameter	Connect															Outputs								Test figure
		Pin 14 to	Pin 33 to	Pin 12 to	Pin 40 to	Pin 37 to	Pin 18 to	Pin 17 to	Pin 20 to	Pin 39 to	Pin 36 to	Pin 13 to	Pin 28 to	Pin 2 to	Pin 8 to	Pin 27 to	Pin 38	Pin 34	Pin 30	Pin 4	Pin 19	Pin 15	Pin 11	Pin 24	
L98	Functional	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	L	L	L	L	L	L	L	L	16
L99	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L100	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L101	"	A	B	A	A	B	A	A	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L102	"	B	B	"	B	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L103	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L104	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L105	"	A	B	B	A	B	B	A	B	B	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L106	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"	"	"	"	"	"	"	"	"	"
L107	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L108	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	L	L	L	L	L	L	L	L	"
L109	"	B	"	"	B	"	"	B	"	"	A	A	A	A	A	"	"	"	"	"	"	"	"	"	"
L110	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
L111	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	H	H	H	H	H	H	H	H	"
L112	"	A	"	"	A	"	"	A	"	"	"	"	"	"	"	"	L	L	L	L	L	L	L	L	"
L113	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L114	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L115	"	B	B	A	B	B	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L116	"	A	B	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L117	"	B	A	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L118	"	A	A	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
L119	"	B	B	B	B	B	B	B	B	B	"	"	"	"	"	"	H	H	H	H	H	H	H	H	"
L120	"	"	"	"	"	"	"	"	"	"	B	B	B	B	"	"	H	H	H	H	H	H	H	H	"
L121	"	"	"	"	"	"	"	"	"	"	B	B	B	B	B	B	L	L	L	L	L	L	L	L	"
L122	"	"	"	"	"	"	"	"	"	"	A	A	A	A	B	B	H	H	H	H	H	H	H	H	"

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. Load all outputs with load as shown on figure 15.
3. Make connections to points A and B in the order listed (left to right) in table XII.
4. Use item MO2 to verify outputs:
   H ≥ 2.4 V.
   L ≤ 0.4 V.

FIGURE 16. Functional test circuit for table XII.

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TABLE XIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	ZB1	Propagation delay time (t_{PHL})	Open	18,37,39,20	40,17	38	3	39	ns	17,18
M2	ZB2	"	"	"	"	34	"	"	"	"
M3	ZB3	"	"	"	"	30	"	"	"	"
M4	ZB4	"	"	"	"	4	"	"	"	"
M5	ZC1	"	"	"	"	19	"	"	"	"
M6	ZC2	"	"	"	"	15	"	"	"	"
M7	ZC3	"	"	"	"	11	"	"	"	"
M8	ZC4	"	"	"	"	24	"	"	"	"
M9	ZB1	"	37,20	18,39	"	38	"	"	"	"
M10	ZB2	"	"	"	"	34	"	"	"	"
M11	ZB3	"	"	"	"	30	"	"	"	"
M12	ZB4	"	"	"	"	4	"	"	"	"
M13	ZC1	"	"	"	"	19	"	"	"	"
M14	ZC2	"	"	"	"	15	"	"	"	"
M15	ZC3	"	"	"	"	11	"	"	"	"
M16	ZC4	"	"	"	"	24	"	"	"	"
M17	ZB1	"	18,39	37,20	"	38	"	"	"	"
M18	ZB2	"	"	"	"	34	"	"	"	"
M19	ZB3	"	"	"	"	30	"	"	"	"
M20	ZB4	"	"	"	"	4	"	"	"	"
M21	ZC1	"	"	"	"	19	"	"	"	"
M22	ZC2	"	"	"	"	15	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M23	ZC3	Propagation delay time (t_{PHL})	18,39	37,20	40,17	11	3	39	ns	17,18
M24	ZC4	"	18,39	37,20	"	24	"	"	"	"
M25	ZB1	"	18,39,37,20	Open	"	38	"	"	"	"
M26	ZB2	"	"	"	"	34	"	"	"	"
M27	ZB3	"	"	"	"	30	"	"	"	"
M28	ZB4	"	"	"	"	4	"	"	"	"
M29	ZC1	"	"	"	"	19	"	"	"	"
M30	ZC2	"	"	"	"	15	"	"	"	"
M31	ZC3	"	"	"	"	11	"	"	"	"
M32	ZC4	"	"	"	"	24	"	"	"	"
M33	ZB1	"	Open	18,37,39,20	"	38	"	"	"	"
M34	ZB2	"	"	"	"	34	"	"	"	"
M35	ZB3	"	"	"	"	30	"	"	"	"
M36	ZB4	"	"	"	"	4	"	"	"	"
M37	ZC1	"	"	"	"	19	"	"	"	"
M38	ZC2	"	"	"	"	15	"	"	"	"
M39	ZC3	"	"	"	"	11	"	"	"	"
M40	ZC4	"	"	"	"	24	"	"	"	"
M41	ZB1	"	37,20	18,39	"	38	"	"	"	"
M42	ZB2	"	"	"	"	34	"	"	"	"
M43	ZB3	"	"	"	"	30	"	"	"	"
M44	ZB4	"	"	"	"	4	"	"	"	"
M45	ZC1	"	"	"	"	19	"	"	"	"
M46	ZC2	"	"	"	"	15	"	"	"	"
M47	ZC3	"	"	"	"	11	"	"	"	"
M48	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M49	ZB1	Propagation delay time (t_{PHL})	18,39	37,20	40,17	38	3	39	ns	17,18
M50	ZB2	"	"	"	"	34	"	"	"	"
M51	ZB3	"	"	"	"	30	"	"	"	"
M52	ZB4	"	"	"	"	4	"	"	"	"
M53	ZC1	"	"	"	"	19	"	"	"	"
M54	ZC2	"	"	"	"	15	"	"	"	"
M55	ZC3	"	"	"	"	11	"	"	"	"
M56	ZC4	"	"	"	"	24	"	"	"	"
M57	ZB1	"	18,39,37,20	Open	"	38	"	"	"	"
M58	ZB2	"	"	"	"	34	"	"	"	"
M59	ZB3	"	"	"	"	30	"	"	"	"
M60	ZB4	"	"	"	"	4	"	"	"	"
M61	ZC1	"	"	"	"	19	"	"	"	"
M62	ZC2	"	"	"	"	15	"	"	"	"
M63	ZC3	"	"	"	"	11	"	"	"	"
M64	ZC4	"	"	"	"	24	"	"	"	"
M65	ZB1	"	Open	18,40,39,17	37,20	38	"	"	"	"
M66	ZB2	"	"	"	"	34	"	"	"	"
M67	ZB3	"	"	"	"	30	"	"	"	"
M68	ZB4	"	"	"	"	4	"	"	"	"
M69	ZC1	"	"	"	"	19	"	"	"	"
M70	ZC2	"	"	"	"	15	"	"	"	"
M71	ZC3	"	"	"	"	11	"	"	"	"
M72	ZC4	"	"	"	"	24	"	"	"	"
M73	ZB1	"	40,17	18,39	"	38	"	"	"	"
M74	ZB2	"	40,17	18,39	"	34	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M75	ZB3	Propagation delay time (t_{PHL})	40,17	18,39	37,20	30	3	39	ns	17,18
M76	ZB4	"	"	"	"	4	"	"	"	"
M77	ZC1	"	"	"	"	19	"	"	"	"
M78	ZC2	"	"	"	"	15	"	"	"	"
M79	ZC3	"	"	"	"	11	"	"	"	"
M80	ZC4	"	"	"	"	24	"	"	"	"
M81	ZB1	"	18,39	40,17	"	38	"	"	"	"
M82	ZB2	"	"	"	"	34	"	"	"	"
M83	ZB3	"	"	"	"	30	"	"	"	"
M84	ZB4	"	"	"	"	4	"	"	"	"
M85	ZC1	"	"	"	"	19	"	"	"	"
M86	ZC2	"	"	"	"	15	"	"	"	"
M87	ZC3	"	"	"	"	11	"	"	"	"
M88	ZC4	"	"	"	"	24	"	"	"	"
M89	ZB1	"	18,39,40,17	Open	"	38	"	"	"	"
M90	ZB2	"	"	"	"	34	"	"	"	"
M91	ZB3	"	"	"	"	30	"	"	"	"
M92	ZB4	"	"	"	"	4	"	"	"	"
M93	ZC1	"	"	"	"	19	"	"	"	"
M94	ZC2	"	"	"	"	15	"	"	"	"
M95	ZC3	"	"	"	"	11	"	"	"	"
M96	ZC4	"	"	"	"	24	"	"	"	"
M97	ZB1	"	Open	18,40,39,17	"	38	"	"	"	"
M98	ZB2	"	"	"	"	34	"	"	"	"
M99	ZB3	"	"	"	"	30	"	"	"	"
M100	ZB4	"	"	"	"	4	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M101	ZC1	Propagation delay time (t_{PHL})	Open	18,40,39,17	37,20	19	3	39	ns	17,18
M102	ZC2	"	"	"	"	15	"	"	"	"
M103	ZC3	"	"	"	"	11	"	"	"	"
M104	ZC4	"	"	"	"	24	"	"	"	"
M105	ZB1	"	40,17	18,39	"	38	"	"	"	"
M106	ZB2	"	"	"	"	34	"	"	"	"
M107	ZB3	"	"	"	"	30	"	"	"	"
M108	ZB4	"	"	"	"	4	"	"	"	"
M109	ZC1	"	"	"	"	19	"	"	"	"
M110	ZC2	"	"	"	"	15	"	"	"	"
M111	ZC3	"	"	"	"	11	"	"	"	"
M112	ZC4	"	"	"	"	24	"	"	"	"
M113	ZB1	"	18,39	40,17	"	38	"	"	"	"
M114	ZB2	"	"	"	"	34	"	"	"	"
M115	ZB3	"	"	"	"	30	"	"	"	"
M116	ZB4	"	"	"	"	4	"	"	"	"
M117	ZC1	"	"	"	"	19	"	"	"	"
M118	ZC2	"	"	"	"	15	"	"	"	"
M119	ZC3	"	"	"	"	11	"	"	"	"
M120	ZC4	"	"	"	"	24	"	"	"	"
M121	ZB1	"	18,39,40,17	Open	"	38	"	"	"	"
M122	ZB2	"	"	"	"	34	"	"	"	"
M123	ZB3	"	"	"	"	30	"	"	"	"
M124	ZB4	"	"	"	"	4	"	"	"	"
M125	ZC1	"	"	"	"	19	"	"	"	"
M126	ZC2	"	"	"	"	15	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M127	ZC3	Propagation delay time (t_{PHL})	18,39,40,17	Open	37,20	11	3	39	ns	17,18
M128	ZC4	"	18,39,40,17	Open	37,20	24	"	"	"	"
M129	ZB1	"	Open	40,17,37,20	18,39	38	"	"	"	"
M130	ZB2	"	"	"	"	34	"	"	"	"
M131	ZB3	"	"	"	"	30	"	"	"	"
M132	ZB4	"	"	"	"	4	"	"	"	"
M133	ZC1	"	"	"	"	19	"	"	"	"
M134	ZC2	"	"	"	"	15	"	"	"	"
M135	ZC3	"	"	"	"	11	"	"	"	"
M136	ZC4	"	"	"	"	24	"	"	"	"
M137	ZB1	"	40,17	37,20	"	38	"	"	"	"
M138	ZB2	"	"	"	"	34	"	"	"	"
M139	ZB3	"	"	"	"	30	"	"	"	"
M140	ZB4	"	"	"	"	4	"	"	"	"
M141	ZC1	"	"	"	"	19	"	"	"	"
M142	ZC2	"	"	"	"	15	"	"	"	"
M143	ZC3	"	"	"	"	11	"	"	"	"
M144	ZC4	"	"	"	"	24	"	"	"	"
M145	ZB1	"	37,20	40,17	"	38	"	"	"	"
M146	ZB2	"	"	"	"	34	"	"	"	"
M147	ZB3	"	"	"	"	30	"	"	"	"
M148	ZB4	"	"	"	"	4	"	"	"	"
M149	ZC1	"	"	"	"	19	"	"	"	"
M150	ZC2	"	"	"	"	15	"	"	"	"
M151	ZC3	"	"	"	"	11	"	"	"	"
M152	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M153	ZB1	Propagation delay time (t_{PHL})	40,17,37,20	Open	18,39	38	3	39	ns	17,18
M154	ZB2	"	"	"	"	34	"	"	"	"
M155	ZB3	"	"	"	"	30	"	"	"	"
M156	ZB4	"	"	"	"	30	"	"	"	"
M157	ZC1	"	"	"	"	19	"	"	"	"
M158	ZC2	"	"	"	"	15	"	"	"	"
M159	ZC3	"	"	"	"	11	"	"	"	"
M160	ZC4	"	"	"	"	24	"	"	"	"
M161	ZB1	"	Open	40,17,37,20	"	38	"	"	"	"
M162	ZB2	"	"	"	"	34	"	"	"	"
M163	ZB3	"	"	"	"	30	"	"	"	"
M164	ZB4	"	"	"	"	4	"	"	"	"
M165	ZC1	"	"	"	"	19	"	"	"	"
M166	ZC2	"	"	"	"	15	"	"	"	"
M167	ZC3	"	"	"	"	11	"	"	"	"
M168	ZC4	"	"	"	"	24	"	"	"	"
M169	ZB1	"	40,17	37,20	"	38	"	"	"	"
M170	ZB2	"	"	"	"	34	"	"	"	"
M171	ZB3	"	"	"	"	30	"	"	"	"
M172	ZB4	"	"	"	"	4	"	"	"	"
M173	ZC1	"	"	"	"	19	"	"	"	"
M174	ZC2	"	"	"	"	15	"	"	"	"
M175	ZC3	"	"	"	"	11	"	"	"	"
M176	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M177	ZB1	Propagation delay time (t_{PHL})	37,20	40,17	18,39	38	3	39	ns	17,18
M178	ZB2	"	"	"	"	34	"	"	"	"
M179	ZB3	"	"	"	"	30	"	"	"	"
M180	ZB4	"	"	"	"	4	"	"	"	"
M181	ZC1	"	"	"	"	19	"	"	"	"
M182	ZC2	"	"	"	"	15	"	"	"	"
M183	ZC3	"	"	"	"	11	"	"	"	"
M184	ZC4	"	"	"	"	24	"	"	"	"
M185	ZB1	"	37,20,40,17	Open	"	38	"	"	"	"
M186	ZB2	"	"	"	"	34	"	"	"	"
M187	ZB3	"	"	"	"	30	"	"	"	"
M188	ZB4	"	"	"	"	4	"	"	"	"
M189	ZC1	"	"	"	"	19	"	"	"	"
M190	ZC2	"	"	"	"	15	"	"	"	"
M191	ZC3	"	"	"	"	11	"	"	"	"
M192	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	ZB1 Propagation delay time (t_{PLH})	Open	18,37,39,20	40,17	38	4	44	ns	17,18
N2	ZB2 "	"	"	"	34	"	"	"	"
N3	ZB3 "	"	"	"	30	"	"	"	"
N4	ZB4 "	"	"	"	4	"	"	"	"
N5	ZC1 "	"	"	"	19	"	"	"	"
N6	ZC2 "	"	"	"	15	"	"	"	"
N7	ZC3 "	"	"	"	11	"	"	"	"
N8	ZC4 "	"	"	"	24	"	"	"	"
N9	ZB1 "	37,20	18,39	"	38	"	"	"	"
N10	ZB2 "	"	"	"	34	"	"	"	"
N11	ZB3 "	"	"	"	30	"	"	"	"
N12	ZB4 "	"	"	"	4	"	"	"	"
N13	ZC1 "	"	"	"	19	"	"	"	"
N14	ZC2 "	"	"	"	15	"	"	"	"
N15	ZC3 "	"	"	"	11	"	"	"	"
N16	ZC4 "	"	"	"	24	"	"	"	"
N17	ZB1 "	18,39	37,20	"	38	"	"	"	"
N18	ZB2 "	"	"	"	34	"	"	"	"
N19	ZB3 "	"	"	"	30	"	"	"	"
N20	ZB4 "	"	"	"	4	"	"	"	"
N21	ZC1 "	"	"	"	19	"	"	"	"
N22	ZC2 "	"	"	"	15	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N23	ZC3	Propagation delay time (t_{PLH})	18,39	37,20	40,17	11	4	44	ns	17,18
N24	ZC4	"	18,39	37,20	"	24	"	"	"	"
N25	ZB1	"	18,39,37,20	Open	"	38	"	"	"	"
N26	ZB2	"	"	"	"	34	"	"	"	"
N27	ZB3	"	"	"	"	30	"	"	"	"
N28	ZB4	"	"	"	"	4	"	"	"	"
N29	ZC1	"	"	"	"	19	"	"	"	"
N30	ZC2	"	"	"	"	15	"	"	"	"
N31	ZC3	"	"	"	"	11	"	"	"	"
N32	ZC4	"	"	"	"	24	"	"	"	"
N33	ZB1	"	Open	18,37,39,20	"	38	"	"	"	"
N34	ZB2	"	"	"	"	34	"	"	"	"
N35	ZB3	"	"	"	"	30	"	"	"	"
N36	ZB4	"	"	"	"	4	"	"	"	"
N37	ZC1	"	"	"	"	19	"	"	"	"
N38	ZC2	"	"	"	"	15	"	"	"	"
N39	ZC3	"	"	"	"	11	"	"	"	"
N40	ZC4	"	"	"	"	24	"	"	"	"
N41	ZB1	"	37,20	18,39	"	38	"	"	"	"
N42	ZB2	"	"	"	"	34	"	"	"	"
N43	ZB3	"	"	"	"	30	"	"	"	"
N44	ZB4	"	"	"	"	4	"	"	"	"
N45	ZC1	"	"	"	"	19	"	"	"	"
N46	ZC2	"	"	"	"	15	"	"	"	"
N47	ZC3	"	"	"	"	11	"	"	"	"
N48	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N49	ZB1	Propagation delay time (t_{PLH})	18,39	37,20	40,17	38	4	44	ns	17,18
N50	ZB2	"	"	"	"	34	"	"	"	"
N51	ZB3	"	"	"	"	30	"	"	"	"
N52	ZB4	"	"	"	"	4	"	"	"	"
N53	ZC1	"	"	"	"	19	"	"	"	"
N54	ZC2	"	"	"	"	15	"	"	"	"
N55	ZC3	"	"	"	"	11	"	"	"	"
N56	ZC4	"	"	"	"		"	"	"	"
N57	ZB1	"	18,39,37,20	Open	"	38	"	"	"	"
N58	ZB2	"	"	"	"	34	"	"	"	"
N59	ZB3	"	"	"	"	30	"	"	"	"
N60	ZB4	"	"	"	"	4	"	"	"	"
N61	ZC1	"	"	"	"	19	"	"	"	"
N62	ZC2	"	"	"	"	15	"	"	"	"
N63	ZC3	"	"	"	"	11	"	"	"	"
N64	ZC4	"	"	"	"	24	"	"	"	"
N65	ZB1	"	Open	18,40,39,17	37,20	38	"	"	"	"
N66	ZB2	"	"	"	"	34	"	"	"	"
N67	ZB3	"	"	"	"	30	"	"	"	"
N68	ZB4	"	"	"	"	4	"	"	"	"
N69	ZC1	"	"	"	"	19	"	"	"	"
N70	ZC2	"	"	"	"	15	"	"	"	"
N71	ZC3	"	"	"	"	11	"	"	"	"
N72	ZC4	"	"	"	"	24	"	"	"	"
N73	ZB1	"	40,17	18,39	"	38	"	"	"	"
N74	ZB2	"	40,17	18,39	"	34	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N75	ZB3	Propagation delay time (t_{PLH})	40,17	18,39	37,20	30	4	44	ns	17,18
N76	ZB4	"	"	"	"	4	"	"	"	"
N77	ZC1	"	"	"	"	19	"	"	"	"
N78	ZC2	"	"	"	"	15	"	"	"	"
N79	ZC3	"	"	"	"	11	"	"	"	"
N80	ZC4	"	"	"	"	24	"	"	"	"
N81	ZB1	"	18,39	40,17	"	38	"	"	"	"
N82	ZB2	"	"	"	"	34	"	"	"	"
N83	ZB3	"	"	"	"	30	"	"	"	"
N84	ZB4	"	"	"	"	4	"	"	"	"
N85	ZC1	"	"	"	"	19	"	"	"	"
N86	ZC2	"	"	"	"	15	"	"	"	"
N87	ZC3	"	"	"	"	11	"	"	"	"
N88	ZC4	"	"	"	"	24	"	"	"	"
N89	ZB1	"	18,39,40,17	Open	"	38	"	"	"	"
N90	ZB2	"	"	"	"	34	"	"	"	"
N91	ZB3	"	"	"	"	30	"	"	"	"
N92	ZB4	"	"	"	"	4	"	"	"	"
N93	ZC1	"	"	"	"	19	"	"	"	"
N94	ZC2	"	"	"	"	15	"	"	"	"
N95	ZC3	"	"	"	"	11	"	"	"	"
N96	ZC4	"	"	"	"	24	"	"	"	"
N97	ZB1	"	Open	18,40,39,17	"	38	"	"	"	"
N98	ZB2	"	"	"	"	34	"	"	"	"
N99	ZB3	"	"	"	"	30	"	"	"	"
N100	ZB4	"	"	"	"	4	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N101	ZC1	Propagation delay time (t_{PLH})	Open	18,40,39,17	37,20	19	4	44	ns	17,18
N102	ZC2	"	"	"	"	15	"	"	"	"
N103	ZC3	"	"	"	"	11	"	"	"	"
N104	ZC4	"	"	"	"	24	"	"	"	"
N105	ZB1	"	40,17	18,39	"	38	"	"	"	"
N106	ZB2	"	"	"	"	34	"	"	"	"
N107	ZB3	"	"	"	"	30	"	"	"	"
N108	ZB4	"	"	"	"	4	"	"	"	"
N109	ZC1	"	"	"	"	19	"	"	"	"
N110	ZC2	"	"	"	"	15	"	"	"	"
N111	ZC3	"	"	"	"	11	"	"	"	"
N112	ZC4	"	"	"	"	24	"	"	"	"
N113	ZB1	"	18,39	40,17	"	38	"	"	"	"
N114	ZB2	"	"	"	"	34	"	"	"	"
N115	ZB3	"	"	"	"	30	"	"	"	"
N116	ZB4	"	"	"	"	4	"	"	"	"
N117	ZC1	"	"	"	"	19	"	"	"	"
N118	ZC2	"	"	"	"	15	"	"	"	"
N119	ZC3	"	"	"	"	11	"	"	"	"
N120	ZC4	"	"	"	"	24	"	"	"	"
N121	ZB1	"	18,39,40,17	Open	"	38	"	"	"	"
N122	ZB2	"	"	"	"	34	"	"	"	"
N123	ZB3	"	"	"	"	30	"	"	"	"
N124	ZB4	"	"	"	"	4	"	"	"	"
N125	ZC1	"	"	"	"	19	"	"	"	"
N126	ZC2	"	"	"	"	15	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N127	ZC3	Propagation delay time (t_{PLH})	18,39,40,17	Open	37,20	11	4	44	ns	17,18
N128	ZC4	"	18,39,40,17	Open	37,20	24	"	"	"	"
N129	ZB1	"	Open	40,17,37,20	18,39	38	"	"	"	"
N130	ZB2	"	"	"	"	34	"	"	"	"
N131	ZB3	"	"	"	"	30	"	"	"	"
N132	ZB4	"	"	"	"	4	"	"	"	"
N133	ZC1	"	"	"	"	19	"	"	"	"
N134	ZC2	"	"	"	"	15	"	"	"	"
N135	ZC3	"	"	"	"	11	"	"	"	"
N136	ZC4	"	"	"	"	24	"	"	"	"
N137	ZB1	"	40,17	37,20	"	38	"	"	"	"
N138	ZB2	"	"	"	"	34	"	"	"	"
N139	ZB3	"	"	"	"	30	"	"	"	"
N140	ZB4	"	"	"	"	4	"	"	"	"
N141	ZC1	"	"	"	"	19	"	"	"	"
N142	ZC2	"	"	"	"	15	"	"	"	"
N143	ZC3	"	"	"	"	11	"	"	"	"
N144	ZC4	"	"	"	"	24	"	"	"	"
N145	ZB1	"	37,20	40,17	"	38	"	"	"	"
N146	ZB2	"	"	"	"	34	"	"	"	"
N147	ZB3	"	"	"	"	30	"	"	"	"
N148	ZB4	"	"	"	"	4	"	"	"	"
N149	ZC1	"	"	"	"	19	"	"	"	"
N150	ZC2	"	"	"	"	15	"	"	"	"
N151	ZC3	"	"	"	"	11	"	"	"	"
N152	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

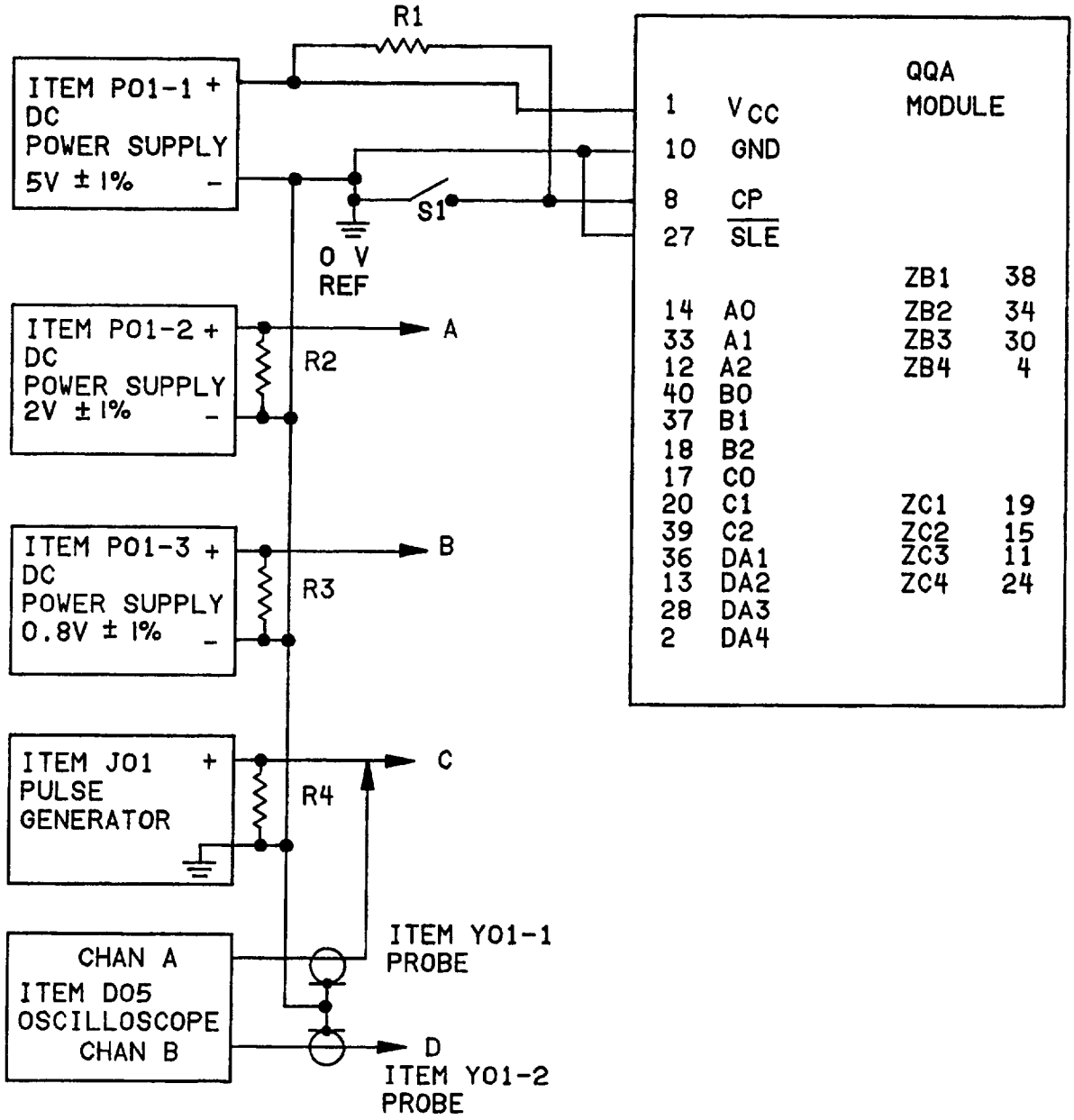
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N153	ZB1	Propagation delay time (t_{PLH})	40,17,37,20	Open	18,39	38	4	44	ns	17,18
N154	ZB2	"	"	"	"	34	"	"	"	"
N155	ZB3	"	"	"	"	30	"	"	"	"
N156	ZB4	"	"	"	"	4	"	"	"	"
N157	ZC1	"	"	"	"	19	"	"	"	"
N158	ZC2	"	"	"	"	15	"	"	"	"
N159	ZC3	"	"	"	"	11	"	"	"	"
N160	ZC4	"	"	"	"	24	"	"	"	"
N161	ZB1	"	Open	40,17,37,20	"	38	"	"	"	"
N162	ZB2	"	"	"	"	34	"	"	"	"
N163	ZB3	"	"	"	"	30	"	"	"	"
N164	ZB4	"	"	"	"	4	"	"	"	"
N165	ZC1	"	"	"	"	19	"	"	"	"
N166	ZC2	"	"	"	"	15	"	"	"	"
N167	ZC3	"	"	"	"	11	"	"	"	"
N168	ZC4	"	"	"	"	24	"	"	"	"
N169	ZB1	"	40,17	37,20	"	38	"	"	"	"
N170	ZB2	"	"	"	"	34	"	"	"	"
N171	ZB3	"	"	"	"	30	"	"	"	"
N172	ZB4	"	"	"	"	4	"	"	"	"
N173	ZC1	"	"	"	"	19	"	"	"	"
N174	ZC2	"	"	"	"	15	"	"	"	"
N175	ZC3	"	"	"	"	11	"	"	"	"
N176	ZC4	"	"	"	"	24	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N177	ZB1	Propagation delay time (t_{PLH})	37,20	40,17	18,39	38	4	44	ns	17,18
N178	ZB2	"	"	"	"	34	"	"	"	"
N179	ZB3	"	"	"	"	30	"	"	"	"
N180	ZB4	"	"	"	"	4	"	"	"	"
N181	ZC1	"	"	"	"	19	"	"	"	"
N182	ZC2	"	"	"	"	15	"	"	"	"
N183	ZC3	"	"	"	"	11	"	"	"	"
N184	ZC4	"	"	"	"	24	"	"	"	"
N185	ZB1	"	37,20,40,17	Open	"	38	"	"	"	"
N186	ZB2	"	"	"	"	34	"	"	"	"
N187	ZB3	"	"	"	"	30	"	"	"	"
N188	ZB4	"	"	"	"	4	"	"	"	"
N189	ZC1	"	"	"	"	19	"	"	"	"
N190	ZC2	"	"	"	"	15	"	"	"	"
N191	ZC3	"	"	"	"	11	"	"	"	"
N192	ZC4	"	"	"	"	24	"	"	"	"

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NOTES:
1. R1 = 1,000Ω ±1%, 1/4 W.
   R2 = R3 = R4 = 51Ω ±5%, 1/4 W.
2. S1 = SPST switch.
3. Close S1 for at least 100 ns and then open. This is to be done at the completion of each step of the setup as specified in notes 4 through 9.

FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}), test circuit for tables XIII and XIV.

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NOTE 4. Setup for test conditions M1-M32 and N33-N64 using figure 16 test circuit.

STEP	Input Conditions							
	A0 14	A1 33	A2 12	DA1 36	DA2 13	DA3 28	DA4 2	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
1	B	B	B	B	B	B	B	
2	B	A	B	B	B	B	B	
3	B	B	A	B	B	B	B	
4	B	A	A	B	B	B	B	
5	A	B	B	A	A	A	A	
6	A	A	B	A	A	A	A	
7	A	B	A	A	A	A	A	
8	A	A	A	A	A	A	A	

NOTE 5. Setup for test conditions M33-M64 and N1-N32 using figure 16 test circuit.

STEP	Input Conditions							
	A0 14	A1 33	A2 12	DA1 36	DA2 13	DA3 28	DA4 2	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
1	B	B	B	A	A	A	A	
2	B	A	B	A	A	A	A	
3	B	B	A	A	A	A	A	
4	B	A	A	A	A	A	A	
5	A	B	B	B	B	B	B	
6	A	A	B	B	B	B	B	
7	A	B	A	B	B	B	B	
8	A	A	A	B	B	B	B	

NOTE 6. Setup for test conditions M65-M96 and N97-N128 using figure 16 test circuit.

STEP	Input Conditions							
	A0 14	A1 33	A2 12	DA1 36	DA2 13	DA3 28	DA4 2	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
1	B	B	B	B	B	B	B	
2	A	B	B	B	B	B	B	
3	B	B	A	B	B	B	B	
4	A	B	A	B	B	B	B	
5	B	A	B	A	A	A	A	
6	A	A	B	A	A	A	A	
7	B	A	A	A	A	A	A	
8	A	A	A	A	A	A	A	

FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIII and XIV - Continued.

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NOTE 7. Setup for test conditions M97-M128 and N65-N96 using figure 16 test circuit.

STEP	Input Conditions							
	A0	A1	A2	DA1	DA2	DA3	DA4	
	14	33	12	36	13	28	2	
1	B	B	B	A	A	A	A	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
2	A	B	B	A	A	A	A	
3	B	B	A	A	A	A	A	
4	A	B	A	A	A	A	A	
5	B	A	B	B	B	B	B	
6	A	A	B	B	B	B	B	
7	B	A	A	B	B	B	B	
8	A	A	A	B	B	B	B	

NOTE 8. Setup for test conditions M129-M160 and N161-N192 using figure 16 test circuit.

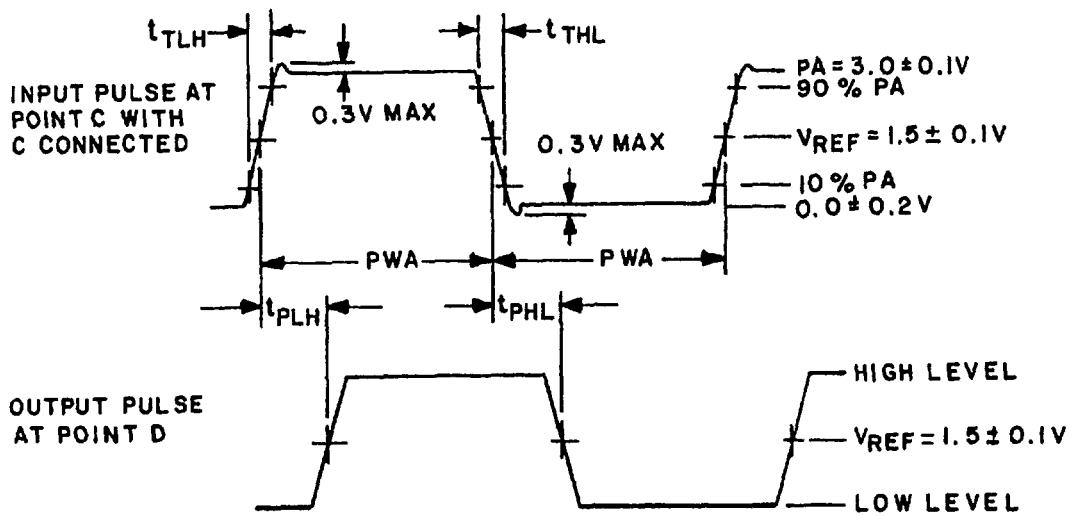
STEP	Input Conditions							
	A0	A1	A2	DA1	DA2	DA3	DA4	
	14	33	12	36	13	28	2	
1	B	B	B	B	B	B	B	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
2	A	B	B	B	B	B	B	
3	B	A	B	B	B	B	B	
4	A	A	B	B	B	B	B	
5	B	B	A	A	A	A	A	
6	A	B	A	A	A	A	A	
7	B	A	A	A	A	A	A	
8	A	A	A	A	A	A	A	

NOTE 9. Setup for test conditions M161-M192 and N129-N160 using figure 16 test circuit.

STEP	Input Conditions							
	A0	A1	A2	DA1	DA2	DA3	DA4	
	14	33	12	36	13	28	2	
1	B	B	B	A	A	A	A	S1 shall be momentarily closed then opened after the input conditions at each step has been established.
2	A	B	B	A	A	A	A	
3	B	A	B	A	A	A	A	
4	A	A	B	A	A	A	A	
5	B	B	A	B	B	B	B	
6	A	B	A	B	B	B	B	
7	B	A	A	B	B	B	B	
8	A	A	A	B	B	B	B	

FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIII and XIV - Continued.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

FIGURE 18. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIII and XIV.

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	DA1 to ZB1 Propagation delay time (t_{PHL})	Open	14,40,17,33,37,20,12, 18,39	36	38	5	54	ns	19,20
P2	DA2 to ZB2 "	"	"	13	34	"	"	"	"
P3	DA3 to ZB3 "	"	"	28	30	"	"	"	"
P4	DA4 to ZB4 "	"	"	2	4	"	"	"	"
P5	DA1 to ZC1 "	"	"	36	19	"	"	"	"
P6	DA2 to ZC2 "	"	"	13	15	"	"	"	"
P7	DA3 to ZC3 "	"	"	28	11	"	"	"	"
P8	DA4 to ZC4 "	"	"	2	24	"	"	"	"
P9	DA1 to ZB1 "	14,40,17	33,37,20,12,18,39	36	38	"	"	"	"
P10	DA2 to ZB2 "	"	"	13	34	"	"	"	"
P11	DA3 to ZB3 "	"	"	28	30	"	"	"	"
P12	DA4 to ZB4 "	"	"	2	4	"	"	"	"
P13	DA1 to ZC1 "	"	"	36	19	"	"	"	"
P14	DA2 to ZC2 "	"	"	13	15	"	"	"	"
P15	DA3 to ZC3 "	"	"	28	11	"	"	"	"
P16	DA4 to ZC4 "	"	"	2	24	"	"	"	"
P17	DA1 to ZB1 "	33,37,20	14,40,17,12,18,39	36	38	"	"	"	"
P18	DA2 to ZB2 "	"	"	13	34	"	"	"	"
P19	DA3 to ZB3 "	"	"	28	30	"	"	"	"
P20	DA4 to ZB4 "	"	"	2	4	"	"	"	"
P21	DA1 to ZC1 "	"	"	36	19	"	"	"	"
P22	DA2 to ZC2 "	"	"	13	15	"	"	"	"
P23	DA3 to ZC3 "	"	"	28	11	"	"	"	"
P24	DA4 to ZC4 "	"	"	2	24	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P25	DA1 to ZB1	Propagation delay time (t_{PHL})	14,40,17, 33,37,20	12,18,39	36	38	5	54	ns	19,20
P26	DA2 to ZB2	"	"	"	13	34	"	"	"	"
P27	DA3 to ZB3	"	"	"	28	30	"	"	"	"
P28	DA4 to ZB4	"	"	"	2	4	"	"	"	"
P29	DA1 to ZC1	"	"	"	36	19	"	"	"	"
P30	DA2 to ZC2	"	"	"	13	15	"	"	"	"
P31	DA3 to ZC3	"	"	"	28	11	"	"	"	"
P32	DA4 to ZC4	"	"	"	2	24	"	"	"	"
P33	DA1 to ZB1	"	12,18,39	14,40,17,33,37,20	36	38	"	"	"	"
P34	DA2 to ZB2	"	"	"	13	34	"	"	"	"
P35	DA3 to ZB3	"	"	"	28	30	"	"	"	"
P36	DA4 to ZB4	"	"	"	2	4	"	"	"	"
P37	DA1 to ZC1	"	"	"	36	19	"	"	"	"
P38	DA2 to ZC2	"	"	"	13	15	"	"	"	"
P39	DA3 to ZC3	"	"	"	28	11	"	"	"	"
P40	DA4 to ZC4	"	"	"	2	24	"	"	"	"
P41	DA1 to ZB1	"	14,40,17, 12,18,39	33,37,20	36	38	"	"	"	"
P42	DA2 to ZB2	"	"	"	13	34	"	"	"	"
P43	DA3 to ZB3	"	"	"	28	30	"	"	"	"
P44	DA4 to ZB4	"	"	"	2	4	"	"	"	"
P45	DA1 to ZC1	"	"	"	36	19	"	"	"	"
P46	DA2 to ZC2	"	"	"	13	15	"	"	"	"
P47	DA3 to ZC3	"	"	"	28	11	"	"	"	"
P48	DA4 to ZC4	"	"	"	2	24	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P49	DA1 to ZB1	Propagation delay time (t_{PHL})	33,37,20, 12,18,39	14,40,17	36	38	5	54	ns	19,20
P50	DA2 to ZB2	"	"	"	13	34	"	"	"	"
P51	DA3 to ZB3	"	"	"	28	30	"	"	"	"
P52	DA4 to ZB4	"	"	"	2	4	"	"	"	"
P53	DA1 to ZC1	"	"	"	36	19	"	"	"	"
P54	DA2 to ZC2	"	"	"	13	15	"	"	"	"
P55	DA3 to ZC3	"	"	"	28	11	"	"	"	"
P56	DA4 to ZC4	"	"	"	2	24	"	"	"	"
P57	DA1 to ZB1	"	14,40,17, 33,37,20, 12,18,39	Open	36	38	"	"	"	"
P58	DA2 to ZB2	"	"	"	13	34	"	"	"	"
P59	DA3 to ZB3	"	"	"	28	30	"	"	"	"
P60	DA4 to ZB4	"	"	"	2	4	"	"	"	"
P61	DA1 to ZC1	"	"	"	36	19	"	"	"	"
P62	DA2 to ZC2	"	"	"	13	15	"	"	"	"
P63	DA3 to ZC3	"	"	"	28	11	"	"	"	"
P64	DA4 to ZC4	"	"	"	2	24	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	DA1 to ZB1 Propagation delay time (t_{PLH})	Open	14,40,17,33,37,20,12, 18,39	36	38	5	49	ns	19,20
Q2	DA2 to ZB2 "	"	"	13	34	"	"	"	"
Q3	DA3 to ZB3 "	"	"	28	30	"	"	"	"
Q4	DA4 to ZB4 "	"	"	2	4	"	"	"	"
Q5	DA1 to ZC1 "	"	"	36	19	"	"	"	"
Q6	DA2 to ZC2 "	"	"	13	15	"	"	"	"
Q7	DA3 to ZC3 "	"	"	28	11	"	"	"	"
Q8	DA4 to ZC4 "	"	"	2	24	"	"	"	"
Q9	DA1 to ZB1 "	14,40,17	33,37,20,12,18,39	36	38	"	"	"	"
Q10	DA2 to ZB2 "	"	"	13	34	"	"	"	"
Q11	DA3 to ZB3 "	"	"	28	30	"	"	"	"
Q12	DA4 to ZB4 "	"	"	2	4	"	"	"	"
Q13	DA1 to ZC1 "	"	"	36	19	"	"	"	"
Q14	DA2 to ZC2 "	"	"	13	15	"	"	"	"
Q15	DA3 to ZC3 "	"	"	28	11	"	"	"	"
Q16	DA4 to ZC4 "	"	"	2	24	"	"	"	"
Q17	DA1 to ZB1 "	33,37,20	14,40,17,12,18,39	36	38	"	"	"	"
Q18	DA2 to ZB2 "	"	"	13	34	"	"	"	"
Q19	DA3 to ZB3 "	"	"	28	30	"	"	"	"
Q20	DA4 to ZB4 "	"	"	2	4	"	"	"	"
Q21	DA1 to ZC1 "	"	"	36	19	"	"	"	"
Q22	DA2 to ZC2 "	"	"	13	15	"	"	"	"
Q23	DA3 to ZC3 "	"	"	28	11	"	"	"	"
Q24	DA4 to ZC4 "	"	"	2	24	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

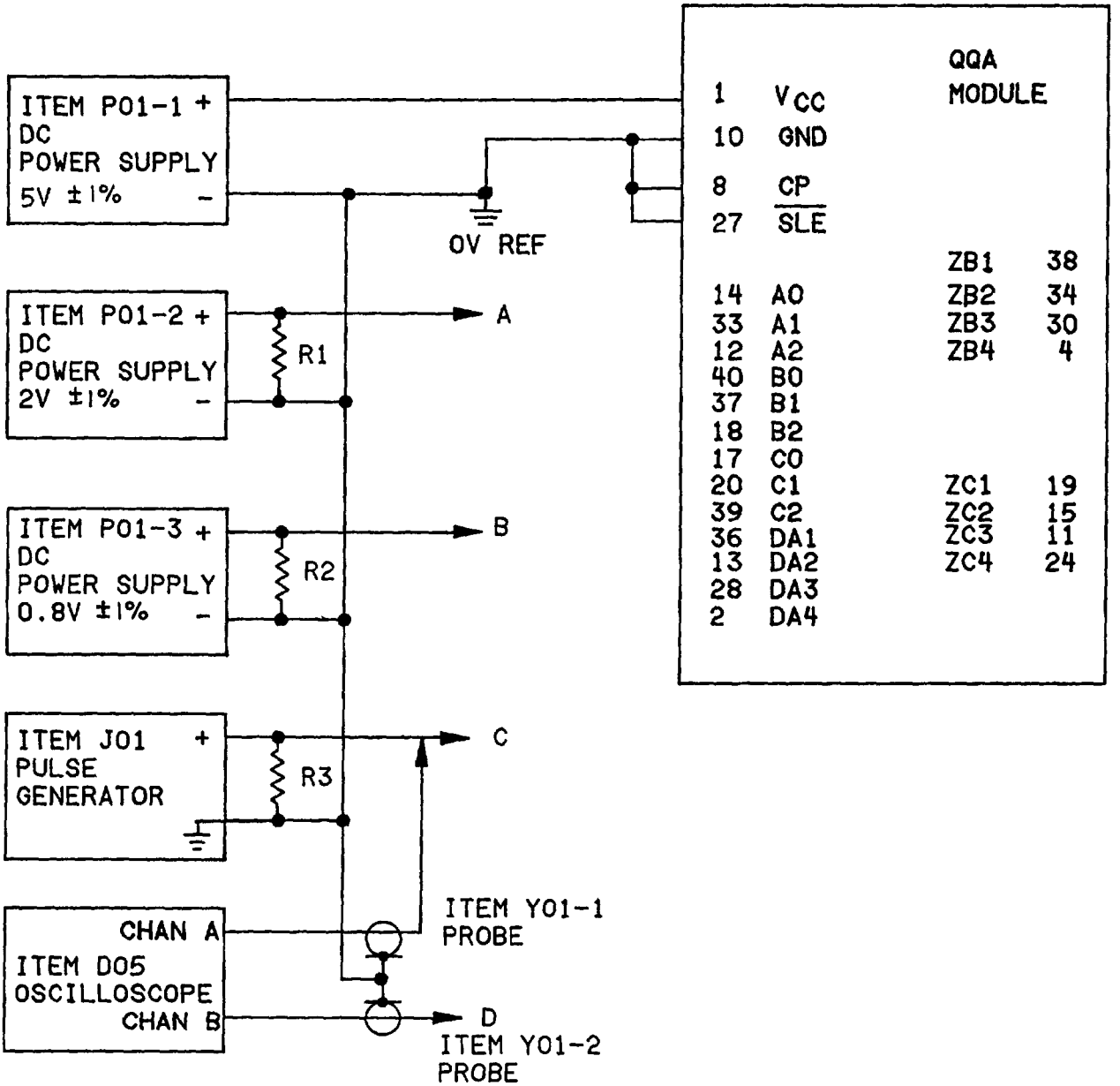
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q25	DA1 to ZB1	Propagation delay time (t_{PLH})	14,40,17, 33,37,20	12,18,39	36	38	5	49	ns	19,20
Q26	DA2 to ZB2	"	"	"	13	34	"	"	"	"
Q27	DA3 to ZB3	"	"	"	28	30	"	"	"	"
Q28	DA4 to ZB4	"	"	"	2	4	"	"	"	"
Q29	DA1 to ZC1	"	"	"	36	19	"	"	"	"
Q30	DA2 to ZC2	"	"	"	13	15	"	"	"	"
Q31	DA3 to ZC3	"	"	"	28	11	"	"	"	"
Q32	DA4 to ZC4	"	"	"	2	24	"	"	"	"
Q33	DA1 to ZB1	"	12,18,39	14,40,17,33,37,20	36	38	"	"	"	"
Q34	DA2 to ZB2	"	"	"	13	34	"	"	"	"
Q35	DA3 to ZB3	"	"	"	28	30	"	"	"	"
Q36	DA4 to ZB4	"	"	"	2	4	"	"	"	"
Q37	DA1 to ZC1	"	"	"	36	19	"	"	"	"
Q38	DA2 to ZC2	"	"	"	13	15	"	"	"	"
Q39	DA3 to ZC3	"	"	"	28	11	"	"	"	"
Q40	DA4 to ZC4	"	"	"	2	24	"	"	"	"
Q41	DA1 to ZB1	"	14,40,17, 12,18,39	33,37,20	36	38	"	"	"	"
Q42	DA2 to ZB2	"	"	"	13	34	"	"	"	"
Q43	DA3 to ZB3	"	"	"	28	30	"	"	"	"
Q44	DA4 to ZB4	"	"	"	2	4	"	"	"	"
Q45	DA1 to ZC1	"	"	"	36	19	"	"	"	"
Q46	DA2 to ZC2	"	"	"	13	15	"	"	"	"
Q47	DA3 to ZC3	"	"	"	28	11	"	"	"	"
Q48	DA4 to ZC4	"	"	"	2	24	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q49	DA1 to ZB1	Propagation delay time (t_{PLH})	33,37,20, 12,18,39	14,40,17	36	38	5	49	ns	19,20
Q50	DA2 to ZB2	"	"	"	13	34	"	"	"	"
Q51	DA3 to ZB3	"	"	"	28	30	"	"	"	"
Q52	DA4 to ZB4	"	"	"	2	4	"	"	"	"
Q53	DA1 to ZC1	"	"	"	36	19	"	"	"	"
Q54	DA2 to ZC2	"	"	"	13	15	"	"	"	"
Q55	DA3 to ZC3	"	"	"	28	11	"	"	"	"
Q56	DA4 to ZC4	"	"	"	2	24	"	"	"	"
Q57	DA1 to ZB1	"	14,40,17, 33,37,20, 12,18,39	Open	36	38	"	"	"	"
Q58	DA2 to ZB2	"	"	"	13	34	"	"	"	"
Q59	DA3 to ZB3	"	"	"	28	30	"	"	"	"
Q60	DA4 to ZB4	"	"	"	2	4	"	"	"	"
Q61	DA1 to ZC1	"	"	"	36	19	"	"	"	"
Q62	DA2 to ZC2	"	"	"	13	15	"	"	"	"
Q63	DA3 to ZC3	"	"	"	28	11	"	"	"	"
Q64	DA4 to ZC4	"	"	"	2	24	"	"	"	"

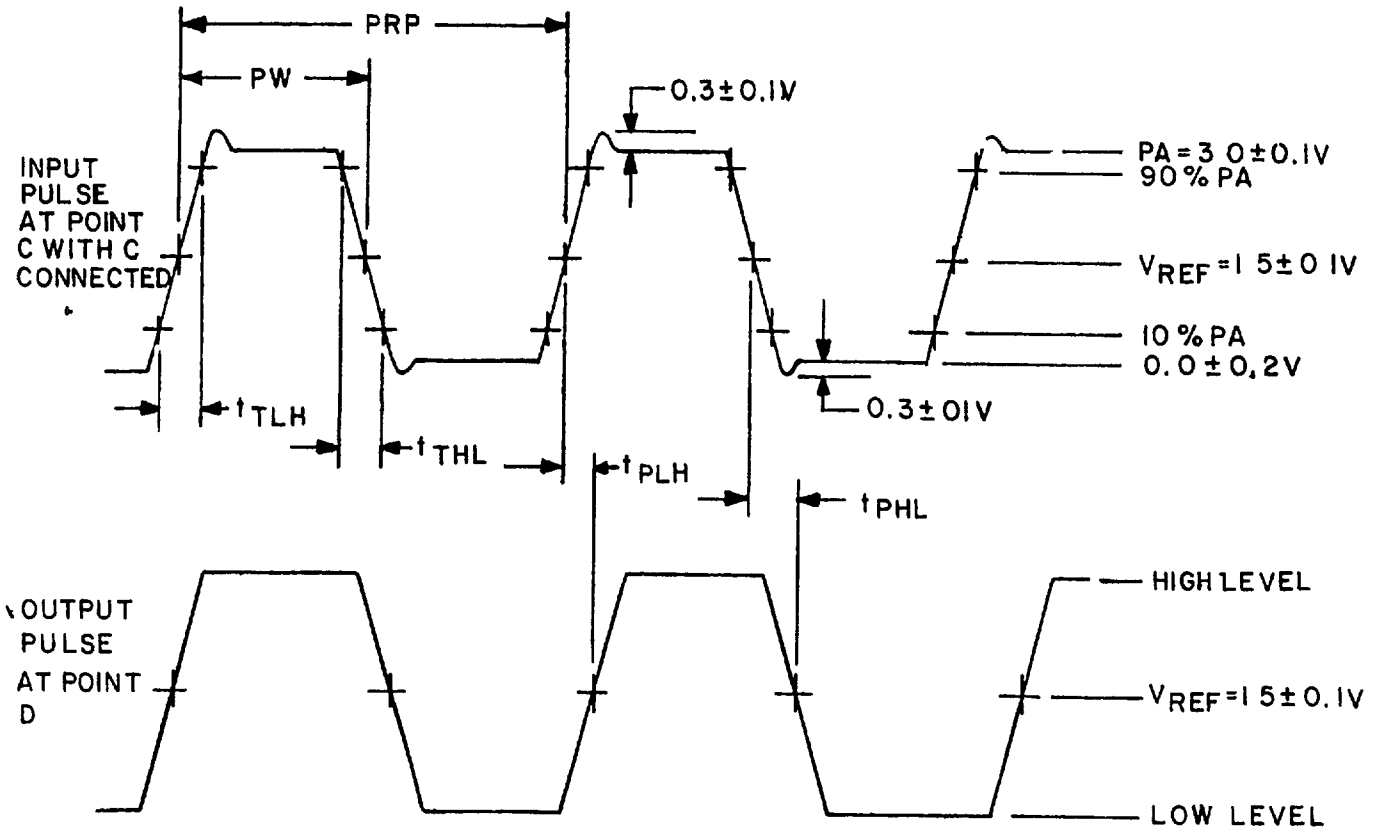
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NOTE: R1 = R2 = R3 = 51Ω ±5%, 1/4 W.

FIGURE 19. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XV and XVI.

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NOTES:
1. Input pulse parameters:
   PRP = 400 ±40 ns.
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. All voltages are referenced to ground.

FIGURE 20. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XV and XVI.

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TABLE XVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	CP to ZB1	Propagation delay time (t_{PHL})	Open	14,40,17,33,37, 20,12,18,39	38	3	34	ns	21,22
R2	CP to ZB2	"	"	"	34	"	"	"	"
R3	CP to ZB3	"	"	"	30	"	"	"	"
R4	CP to ZB4	"	"	"	4	"	"	"	"
R5	CP to ZC1	"	"	"	19	"	"	"	"
R6	CP to ZC2	"	"	"	15	"	"	"	"
R7	CP to ZC3	"	"	"	11	"	"	"	"
R8	CP to ZC4	"	"	"	24	"	"	"	"
R9	CP to ZB1	"	14,40,17	33,37,20,12,18,39	38	"	"	"	"
R10	CP to ZB2	"	"	"	34	"	"	"	"
R11	CP to ZB3	"	"	"	30	"	"	"	"
R12	CP to ZB4	"	"	"	4	"	"	"	"
R13	CP to ZC1	"	"	"	19	"	"	"	"
R14	CP to ZC2	"	"	"	15	"	"	"	"
R15	CP to ZC3	"	"	"	11	"	"	"	"
R16	CP to ZC4	"	"	"	24	"	"	"	"
R17	CP to ZB1	"	33,37,20	14,40,17,12,18,39	38	"	"	"	"
R18	CP to ZB2	"	"	"	34	"	"	"	"
R19	CP to ZB3	"	"	"	30	"	"	"	"
R20	CP to ZB4	"	"	"	4	"	"	"	"
R21	CP to ZC1	"	"	"	19	"	"	"	"
R22	CP to ZC2	"	"	"	15	"	"	"	"
R23	CP to ZC3	"	"	"	11	"	"	"	"
R24	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R25	CP to ZB1	Propagation delay time (t_{PHL})	14,40,17,33,37,20	12,18,39	38	3	34	ns	21,22
R26	CP to ZB2	"	"	"	34	"	"	"	"
R27	CP to ZB3	"	"	"	30	"	"	"	"
R28	CP to ZB4	"	"	"	4	"	"	"	"
R29	CP to ZC1	"	"	"	19	"	"	"	"
R30	CP to ZC2	"	"	"	15	"	"	"	"
R31	CP to ZC3	"	"	"	11	"	"	"	"
R32	CP to ZC4	"	"	"	24	"	"	"	"
R33	CP to ZB1	"	12,18,39	14,40,17,33,37,20	38	"	"	"	"
R34	CP to ZB2	"	"	"	34	"	"	"	"
R35	CP to ZB3	"	"	"	30	"	"	"	"
R36	CP to ZB4	"	"	"	4	"	"	"	"
R37	CP to ZC1	"	"	"	19	"	"	"	"
R38	CP to ZC2	"	"	"	15	"	"	"	"
R39	CP to ZC3	"	"	"	11	"	"	"	"
R40	CP to ZC4	"	"	"	24	"	"	"	"
R41	CP to ZB1	"	14,40,17,12,18,39	33,37,20	38	"	"	"	"
R42	CP to ZB2	"	"	"	34	"	"	"	"
R43	CP to ZB3	"	"	"	30	"	"	"	"
R44	CP to ZB4	"	"	"	4	"	"	"	"
R45	CP to ZC1	"	"	"	19	"	"	"	"
R46	CP to ZC2	"	"	"	15	"	"	"	"
R47	CP to ZC3	"	"	"	11	"	"	"	"
R48	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R49	CP to ZB1	Propagation delay time (t_{PHL})	33,37,20,12,18,39	14,40,17	38	3	34	ns	21,22
R50	CP to ZB2	"	"	"	34	"	"	"	"
R51	CP to ZB3	"	"	"	30	"	"	"	"
R52	CP to ZB4	"	"	"	4	"	"	"	"
R53	CP to ZC1	"	"	"	19	"	"	"	"
R54	CP to ZC2	"	"	"	15	"	"	"	"
R55	CP to ZC3	"	"	"	11	"	"	"	"
R56	CP to ZC4	"	"	"	24	"	"	"	"
R57	CP to ZB1	"	14,40,17,33,37, 20,12,18,39	Open	38	"	"	"	"
R58	CP to ZB2	"	"	"	34	"	"	"	"
R59	CP to ZB3	"	"	"	30	"	"	"	"
R60	CP to ZB4	"	"	"	4	"	"	"	"
R61	CP to ZC1	"	"	"	19	"	"	"	"
R62	CP to ZC2	"	"	"	15	"	"	"	"
R63	CP to ZC3	"	"	"	11	"	"	"	"
R64	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	CP to ZB1	Propagation delay time (t_{PLH})	Open	14,40,17,33,37, 20,12,18,39	38	3	39	ns	21,22
S2	CP to ZB2	"	"	"	34	"	"	"	"
S3	CP to ZB3	"	"	"	30	"	"	"	"
S4	CP to ZB4	"	"	"	4	"	"	"	"
S5	CP to ZC1	"	"	"	19	"	"	"	"
S6	CP to ZC2	"	"	"	15	"	"	"	"
S7	CP to ZC3	"	"	"	11	"	"	"	"
S8	CP to ZC4	"	"	"	24	"	"	"	"
S9	CP to ZB1	"	14,40,17	33,37,20,12,18,39	38	"	"	"	"
S10	CP to ZB2	"	"	"	34	"	"	"	"
S11	CP to ZB3	"	"	"	30	"	"	"	"
S12	CP to ZB4	"	"	"	4	"	"	"	"
S13	CP to ZC1	"	"	"	19	"	"	"	"
S14	CP to ZC2	"	"	"	15	"	"	"	"
S15	CP to ZC3	"	"	"	11	"	"	"	"
S16	CP to ZC4	"	"	"	24	"	"	"	"
S17	CP to ZB1	"	33,37,20	14,40,17,12,18,39	38	"	"	"	"
S18	CP to ZB2	"	"	"	34	"	"	"	"
S19	CP to ZB3	"	"	"	30	"	"	"	"
S20	CP to ZB4	"	"	"	4	"	"	"	"
S21	CP to ZC1	"	"	"	19	"	"	"	"
S22	CP to ZC2	"	"	"	15	"	"	"	"
S23	CP to ZC3	"	"	"	11	"	"	"	"
S24	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S25	CP to ZB1	Propagation delay time (t_{PLH})	14,40,17,33,37,20	12,18,39	38	3	39	ns	21,22
S26	CP to ZB2	"	"	"	34	"	"	"	"
S27	CP to ZB3	"	"	"	30	"	"	"	"
S28	CP to ZB4	"	"	"	4	"	"	"	"
S29	CP to ZC1	"	"	"	19	"	"	"	"
S30	CP to ZC2	"	"	"	15	"	"	"	"
S31	CP to ZC3	"	"	"	11	"	"	"	"
S32	CP to ZC4	"	"	"	24	"	"	"	"
S33	CP to ZB1	"	12,18,39	14,40,17,33,37,20	38	"	"	"	"
S34	CP to ZB2	"	"	"	34	"	"	"	"
S35	CP to ZB3	"	"	"	30	"	"	"	"
S36	CP to ZB4	"	"	"	4	"	"	"	"
S37	CP to ZC1	"	"	"	19	"	"	"	"
S38	CP to ZC2	"	"	"	15	"	"	"	"
S39	CP to ZC3	"	"	"	11	"	"	"	"
S40	CP to ZC4	"	"	"	24	"	"	"	"
S41	CP to ZB1	"	14,40,17,12,18,39	33,37,20	38	"	"	"	"
S42	CP to ZB2	"	"	"	34	"	"	"	"
S43	CP to ZB3	"	"	"	30	"	"	"	"
S44	CP to ZB4	"	"	"	4	"	"	"	"
S45	CP to ZC1	"	"	"	19	"	"	"	"
S46	CP to ZC2	"	"	"	15	"	"	"	"
S47	CP to ZC3	"	"	"	11	"	"	"	"
S48	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S49	CP to ZB1	Propagation delay time (t_{PLH})	33,37,20,12,18,39	14,40,17	38	3	39	ns	21,22
S50	CP to ZB2	"	"	"	34	"	"	"	"
S51	CP to ZB3	"	"	"	30	"	"	"	"
S52	CP to ZB4	"	"	"	4	"	"	"	"
S53	CP to ZC1	"	"	"	19	"	"	"	"
S54	CP to ZC2	"	"	"	15	"	"	"	"
S55	CP to ZC3	"	"	"	11	"	"	"	"
S56	CP to ZC4	"	"	"	24	"	"	"	"
S57	CP to ZB1	"	14,40,17,33,37, 20,12,18,39	Open	38	"	"	"	"
S58	CP to ZB2	"	"	"	34	"	"	"	"
S59	CP to ZB3	"	"	"	30	"	"	"	"
S60	CP to ZB4	"	"	"	4	"	"	"	"
S61	CP to ZC1	"	"	"	19	"	"	"	"
S62	CP to ZC2	"	"	"	15	"	"	"	"
S63	CP to ZC3	"	"	"	11	"	"	"	"
S64	CP to ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	ZB1	Minimum setup and hold time (t_{SL})	Open	14,40,17,33,37, 20,12,18,39	38	155	245	ns	21,23
T2	ZB2	"	"	"	34	"	"	"	"
T3	ZB3	"	"	"	30	"	"	"	"
T4	ZB4	"	"	"	4	"	"	"	"
T5	ZC1	"	"	"	19	"	"	"	"
T6	ZC2	"	"	"	15	"	"	"	"
T7	ZC3	"	"	"	11	"	"	"	"
T8	ZC4	"	"	"	24	"	"	"	"
T9	ZB1	"	14,40,17	33,37,20,12,18,39	38	"	"	"	"
T10	ZB2	"	"	"	34	"	"	"	"
T11	ZB3	"	"	"	30	"	"	"	"
T12	ZB4	"	"	"	4	"	"	"	"
T13	ZC1	"	"	"	19	"	"	"	"
T14	ZC2	"	"	"	15	"	"	"	"
T15	ZC3	"	"	"	11	"	"	"	"
T16	ZC4	"	"	"	24	"	"	"	"
T17	ZB1	"	33,37,20	14,40,17,12,18,39	38	"	"	"	"
T18	ZB2	"	"	"	34	"	"	"	"
T19	ZB3	"	"	"	30	"	"	"	"
T20	ZB4	"	"	"	4	"	"	"	"
T21	ZC1	"	"	"	19	"	"	"	"
T22	ZC2	"	"	"	15	"	"	"	"
T23	ZC3	"	"	"	11	"	"	"	"
T24	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T25	ZB1	Minimum setup and hold time (t_{SL})	14,40,17,33,37,20	12,18,39	38	155	245	ns	21,23
T26	ZB2	"	"	"	34	"	"	"	"
T27	ZB3	"	"	"	30	"	"	"	"
T28	ZB4	"	"	"	4	"	"	"	"
T29	ZC1	"	"	"	19	"	"	"	"
T30	ZC2	"	"	"	15	"	"	"	"
T31	ZC3	"	"	"	11	"	"	"	"
T32	ZC4	"	"	"	24	"	"	"	"
T33	ZB1	"	12,18,39	14,40,17,33,37,20	38	"	"	"	"
T34	ZB2	"	"	"	34	"	"	"	"
T35	ZB3	"	"	"	30	"	"	"	"
T36	ZB4	"	"	"	4	"	"	"	"
T37	ZC1	"	"	"	19	"	"	"	"
T38	ZC2	"	"	"	15	"	"	"	"
T39	ZC3	"	"	"	11	"	"	"	"
T40	ZC4	"	"	"	24	"	"	"	"
T41	ZB1	"	12,18,39,14,40,17	33,37,20	38	"	"	"	"
T42	ZB2	"	"	"	34	"	"	"	"
T43	ZB3	"	"	"	30	"	"	"	"
T44	ZB4	"	"	"	4	"	"	"	"
T45	ZC1	"	"	"	19	"	"	"	"
T46	ZC2	"	"	"	15	"	"	"	"
T47	ZC3	"	"	"	11	"	"	"	"
T48	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T49	ZB1	Minimum setup and hold time (t_{SL})	33,37,20,12,18,39	14,40,17	38	155	245	ns	21,23
T50	ZB2	"	"	"	34	"	"	"	"
T51	ZB3	"	"	"	30	"	"	"	"
T52	ZB4	"	"	"	4	"	"	"	"
T53	ZC1	"	"	"	19	"	"	"	"
T54	ZC2	"	"	"	15	"	"	"	"
T55	ZC3	"	"	"	11	"	"	"	"
T56	ZC4	"	"	"	24	"	"	"	"
T57	ZB1	"	14,40,17,33,37, 20,12,18,39	Open	38	"	"	"	"
T58	ZB2	"	"	"	34	"	"	"	"
T59	ZB3	"	"	"	30	"	"	"	"
T60	ZB4	"	"	"	4	"	"	"	"
T61	ZC1	"	"	"	19	"	"	"	"
T62	ZC2	"	"	"	15	"	"	"	"
T63	ZC3	"	"	"	11	"	"	"	"
T64	ZC4	"	"	"	24	"	"	"	"
T65	ZB1	Minimum setup and hold time (t_{SH})	Open	14,40,17,33,37, 20,12,18,39	38	"	"	"	21,24
T66	ZB2	"	"	"	34	"	"	"	"
T67	ZB3	"	"	"	30	"	"	"	"
T68	ZB4	"	"	"	4	"	"	"	"
T69	ZC1	"	"	"	19	"	"	"	"
T70	ZC2	"	"	"	15	"	"	"	"
T71	ZC3	"	"	"	11	"	"	"	"
T72	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T73	ZB1	Minimum setup and hold time (t_{SH})	14,40,17	33,37,20,12,18,39	38	155	245	ns	21,24
T74	ZB2	"	"	"	34	"	"	"	"
T75	ZB3	"	"	"	30	"	"	"	"
T76	ZB4	"	"	"	4	"	"	"	"
T77	ZC1	"	"	"	19	"	"	"	"
T78	ZC2	"	"	"	15	"	"	"	"
T79	ZC3	"	"	"	11	"	"	"	"
T80	ZC4	"	"	"	24	"	"	"	"
T81	ZB1	"	33,37,20	14,40,17,12,18,39	38	"	"	"	"
T82	ZB2	"	"	"	34	"	"	"	"
T83	ZB3	"	"	"	30	"	"	"	"
T84	ZB4	"	"	"	4	"	"	"	"
T85	ZC1	"	"	"	19	"	"	"	"
T86	ZC2	"	"	"	15	"	"	"	"
T87	ZC3	"	"	"	11	"	"	"	"
T88	ZC4	"	"	"	24	"	"	"	"
T89	ZB1	"	14,40,17,33,37,20	12,18,39	38	"	"	"	"
T90	ZB2	"	"	"	34	"	"	"	"
T91	ZB3	"	"	"	30	"	"	"	"
T92	ZB4	"	"	"	4	"	"	"	"
T93	ZC1	"	"	"	19	"	"	"	"
T94	ZC2	"	"	"	15	"	"	"	"
T95	ZC3	"	"	"	11	"	"	"	"
T96	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T97	ZB1	Minimum setup and hold time (t_{SH})	12,18,39	14,40,17,33,37,20	38	155	245	ns	21,24
T98	ZB2	"	"	"	34	"	"	"	"
T99	ZB3	"	"	"	30	"	"	"	"
T100	ZB4	"	"	"	4	"	"	"	"
T101	ZC1	"	"	"	19	"	"	"	"
T102	ZC2	"	"	"	15	"	"	"	"
T103	ZC3	"	"	"	11	"	"	"	"
T104	ZC4	"	"	"	24	"	"	"	"
T105	ZB1	"	12,18,39,14,40,17	33,37,20	38	"	"	"	"
T106	ZB2	"	"	"	34	"	"	"	"
T107	ZB3	"	"	"	30	"	"	"	"
T108	ZB4	"	"	"	4	"	"	"	"
T109	ZC1	"	"	"	19	"	"	"	"
T110	ZC2	"	"	"	15	"	"	"	"
T111	ZC3	"	"	"	11	"	"	"	"
T112	ZC4	"	"	"	24	"	"	"	"
T113	ZB1	"	33,37,20,12,18,39	14,40,17	38	"	"	"	"
T114	ZB2	"	"	"	34	"	"	"	"
T115	ZB3	"	"	"	30	"	"	"	"
T116	ZB4	"	"	"	4	"	"	"	"
T117	ZC1	"	"	"	19	"	"	"	"
T118	ZC2	"	"	"	15	"	"	"	"
T119	ZC3	"	"	"	11	"	"	"	"
T120	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T121	ZB1	Minimum setup and hold time (t_{SH})	14,40,17,33,37, 20,12,18,39	Open	38	155	245	ns	21,24
T122	ZB2	"	"	"	34	"	"	"	"
T123	ZB3	"	"	"	30	"	"	"	"
T124	ZB4	"	"	"	4	"	"	"	"
T125	ZC1	"	"	"	19	"	"	"	"
T126	ZC2	"	"	"	15	"	"	"	"
T127	ZC3	"	"	"	11	"	"	"	"
T128	ZC4	"	"	"	24	"	"	"	"
T129	ZB1	Minimum setup and hold time (t_{HL})	Open	14,40,17,33,37, 20,12,18,39	38	"	"	"	21,25
T130	ZB2	"	"	"	34	"	"	"	"
T131	ZB3	"	"	"	30	"	"	"	"
T132	ZB4	"	"	"	4	"	"	"	"
T133	ZC1	"	"	"	19	"	"	"	"
T134	ZC2	"	"	"	15	"	"	"	"
T135	ZC3	"	"	"	11	"	"	"	"
T136	ZC4	"	"	"	24	"	"	"	"
T137	ZB1	"	14,40,17	33,37,20,12,18,39	38	"	"	"	"
T138	ZB2	"	"	"	34	"	"	"	"
T139	ZB3	"	"	"	30	"	"	"	"
T140	ZB4	"	"	"	4	"	"	"	"
T141	ZC1	"	"	"	19	"	"	"	"
T142	ZC2	"	"	"	15	"	"	"	"
T143	ZC3	"	"	"	11	"	"	"	"
T144	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T145	ZB1	Minimum setup and hold time (t_{HL})	33,37,20	14,40,17,12,18,39	38	155	245	ns	21,25
T146	ZB2	"	"	"	34	"	"	"	"
T147	ZB3	"	"	"	30	"	"	"	"
T148	ZB4	"	"	"	4	"	"	"	"
T149	ZC1	"	"	"	19	"	"	"	"
T150	ZC2	"	"	"	15	"	"	"	"
T151	ZC3	"	"	"	11	"	"	"	"
T152	ZC4	"	"	"	24	"	"	"	"
T153	ZB1	"	14,40,17,33,37,20	12,18,39	38	"	"	"	"
T154	ZB2	"	"	"	34	"	"	"	"
T155	ZB3	"	"	"	30	"	"	"	"
T156	ZB4	"	"	"	4	"	"	"	"
T157	ZC1	"	"	"	19	"	"	"	"
T158	ZC2	"	"	"	15	"	"	"	"
T159	ZC3	"	"	"	11	"	"	"	"
T160	ZC4	"	"	"	24	"	"	"	"
T161	ZB1	"	12,18,39	14,40,17,33,37,20	38	"	"	"	"
T162	ZB2	"	"	"	34	"	"	"	"
T163	ZB3	"	"	"	30	"	"	"	"
T164	ZB4	"	"	"	4	"	"	"	"
T165	ZC1	"	"	"	19	"	"	"	"
T166	ZC2	"	"	"	15	"	"	"	"
T167	ZC3	"	"	"	11	"	"	"	"
T168	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T169	ZB1	Minimum setup and hold time (t_{HL})	12,18,39,14,40,17	33,37,20	38	155	245	ns	21,25
T170	ZB2	"	"	"	34	"	"	"	"
T171	ZB3	"	"	"	30	"	"	"	"
T172	ZB4	"	"	"	4	"	"	"	"
T173	ZC1	"	"	"	19	"	"	"	"
T174	ZC2	"	"	"	15	"	"	"	"
T175	ZC3	"	"	"	11	"	"	"	"
T176	ZC4	"	"	"	24	"	"	"	"
T177	ZB1	"	33,37,20,12,18,39	14,40,17	38	"	"	"	"
T178	ZB2	"	"	"	34	"	"	"	"
T179	ZB3	"	"	"	30	"	"	"	"
T180	ZB4	"	"	"	4	"	"	"	"
T181	ZC1	"	"	"	19	"	"	"	"
T182	ZC2	"	"	"	15	"	"	"	"
T183	ZC3	"	"	"	11	"	"	"	"
T184	ZC4	"	"	"	24	"	"	"	"
T185	ZB1	"	14,40,17,33,37, 20,12,18,39	Open	38	"	"	"	"
T186	ZB2	"	"	"	34	"	"	"	"
T187	ZB3	"	"	"	30	"	"	"	"
T188	ZB4	"	"	"	4	"	"	"	"
T189	ZC1	"	"	"	19	"	"	"	"
T190	ZC2	"	"	"	15	"	"	"	"
T191	ZC3	"	"	"	11	"	"	"	"
T192	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T193	ZB1	Minimum setup and hold time (t_{HH})	Open	14,40,17,33,37, 20,12,18,39	38	155	245	ns	21,26
T194	ZB2	"	"	"	34	"	"	"	"
T195	ZB3	"	"	"	30	"	"	"	"
T196	ZB4	"	"	"	4	"	"	"	"
T197	ZC1	"	"	"	19	"	"	"	"
T198	ZC2	"	"	"	15	"	"	"	"
T199	ZC3	"	"	"	11	"	"	"	"
T200	ZC4	"	"	"	24	"	"	"	"
T201	ZB1	"	14,40,17	33,37,20,12,18,39	38	"	"	"	"
T202	ZB2	"	"	"	34	"	"	"	"
T203	ZB3	"	"	"	30	"	"	"	"
T204	ZB4	"	"	"	4	"	"	"	"
T205	ZC1	"	"	"	19	"	"	"	"
T206	ZC2	"	"	"	15	"	"	"	"
T207	ZC3	"	"	"	11	"	"	"	"
T208	ZC4	"	"	"	24	"	"	"	"
T209	ZB1	"	33,37,20	14,40,17,12,18,39	38	"	"	"	"
T210	ZB2	"	"	"	34	"	"	"	"
T211	ZB3	"	"	"	30	"	"	"	"
T212	ZB4	"	"	"	4	"	"	"	"
T213	ZC1	"	"	"	19	"	"	"	"
T214	ZC2	"	"	"	15	"	"	"	"
T215	ZC3	"	"	"	11	"	"	"	"
T216	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

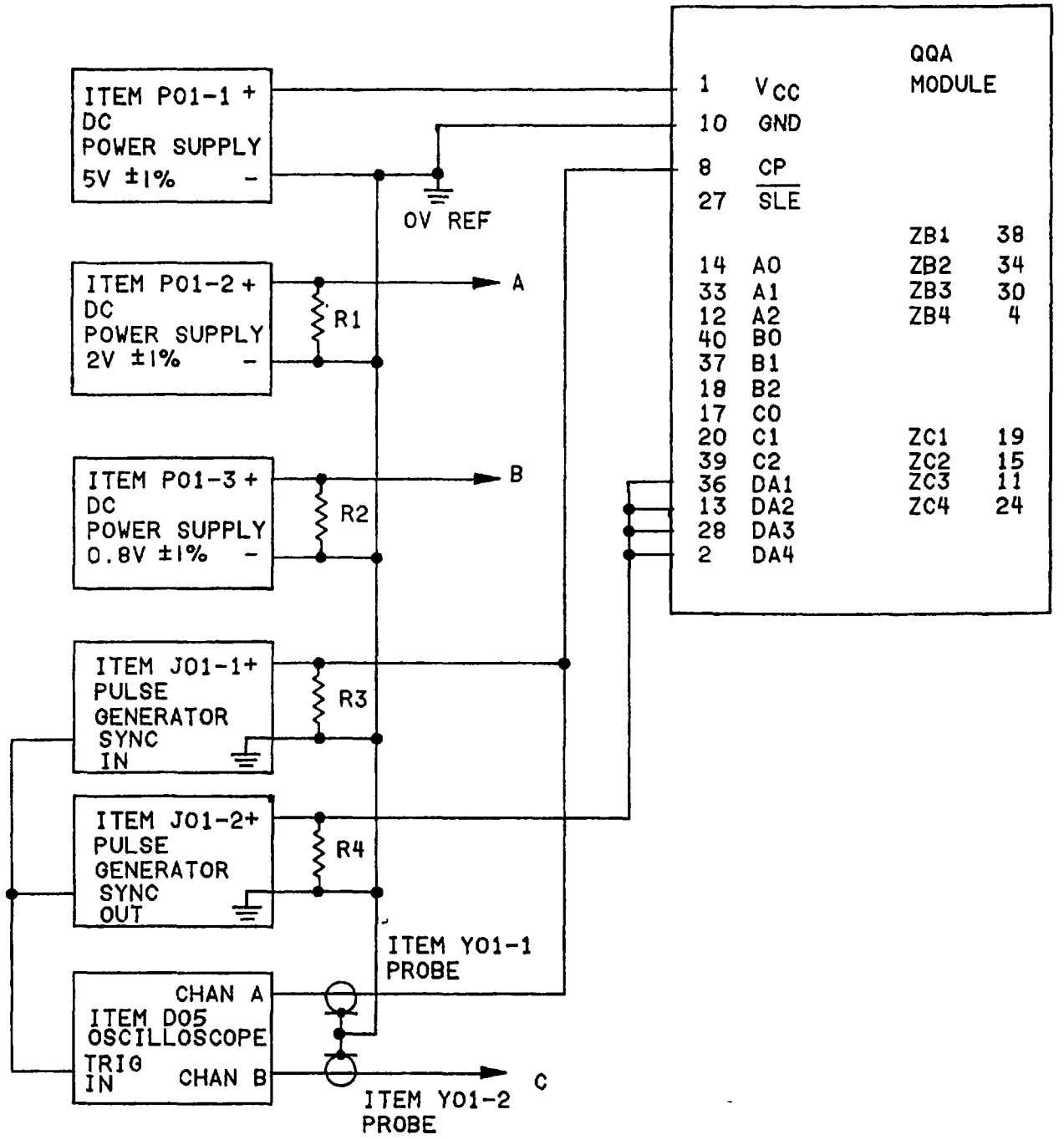
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T217	ZB1	Minimum setup and hold time (t_{HH})	14,40,17,33,37,20	12,18,39	38	155	245	ns	21,26
T218	ZB2	"	"	"	34	"	"	"	"
T219	ZB3	"	"	"	30	"	"	"	"
T220	ZB4	"	"	"	4	"	"	"	"
T221	ZC1	"	"	"	19	"	"	"	"
T222	ZC2	"	"	"	15	"	"	"	"
T223	ZC3	"	"	"	11	"	"	"	"
T224	ZC4	"	"	"	24	"	"	"	"
T225	ZB1	"	12,18,39	14,40,17,33,37,20	38	"	"	"	"
T226	ZB2	"	"	"	34	"	"	"	"
T227	ZB3	"	"	"	30	"	"	"	"
T228	ZB4	"	"	"	4	"	"	"	"
T229	ZC1	"	"	"	19	"	"	"	"
T230	ZC2	"	"	"	15	"	"	"	"
T231	ZC3	"	"	"	11	"	"	"	"
T232	ZC4	"	"	"	24	"	"	"	"
T233	ZB1	"	12,18,39,14,40,17	33,37,20	38	"	"	"	"
T234	ZB2	"	"	"	34	"	"	"	"
T235	ZB3	"	"	"	30	"	"	"	"
T236	ZB4	"	"	"	4	"	"	"	"
T237	ZC1	"	"	"	19	"	"	"	"
T238	ZC2	"	"	"	15	"	"	"	"
T239	ZC3	"	"	"	11	"	"	"	"
T240	ZC4	"	"	"	24	"	"	"	"

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TABLE XIX. Minimum setup and hold time (t_{SL}, t_{SH}, t_{HL} and t_{HH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T241	ZB1	Minimum setup and hold time (t_{HH})	33,37,20,12,18,39	14,40,17	38	155	245	ns	21,26
T242	ZB2	"	"	"	34	"	"	"	"
T243	ZB3	"	"	"	30	"	"	"	"
T244	ZB4	"	"	"	4	"	"	"	"
T245	ZC1	"	"	"	19	"	"	"	"
T246	ZC2	"	"	"	15	"	"	"	"
T247	ZC3	"	"	"	11	"	"	"	"
T248	ZC4	"	"	"	24	"	"	"	"
T249	ZB1	"	14,40,17,33,37, 20,12,18,39	Open	38	"	"	"	"
T250	ZB2	"	"	"	34	"	"	"	"
T251	ZB3	"	"	"	30	"	"	"	"
T252	ZB4	"	"	"	4	"	"	"	"
T253	ZC1	"	"	"	19	"	"	"	"
T254	ZC2	"	"	"	15	"	"	"	"
T255	ZC3	"	"	"	11	"	"	"	"
T256	ZC4	"	"	"	24	"	"	"	"

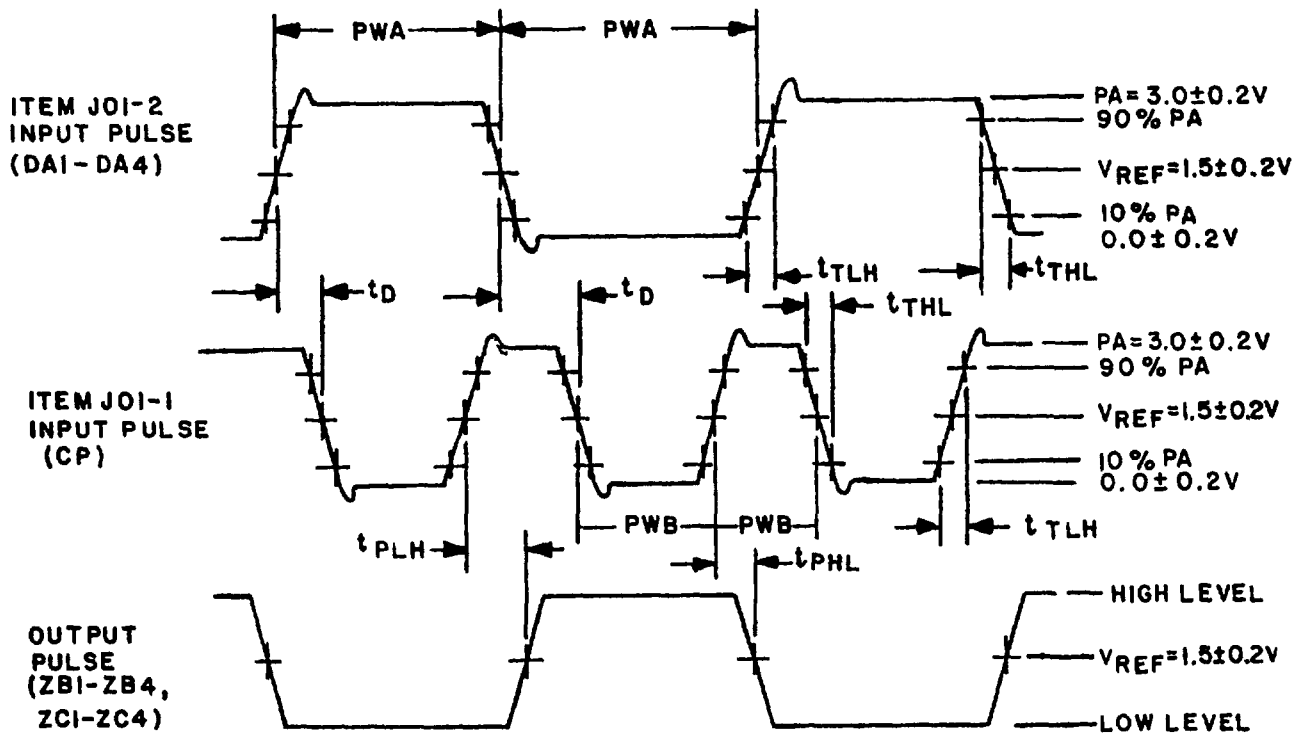
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NOTE: R1-R4 = 51Ω ±5%, 1/4 W.

FIGURE 21. Propagation delay time (t_{PHL} and t_{PLH}) and minimum setup and hold time (t_{SL}, t_{SH}, t_{THL} and t_{HH}) test circuit for tables XVII, XVIII and XIX.

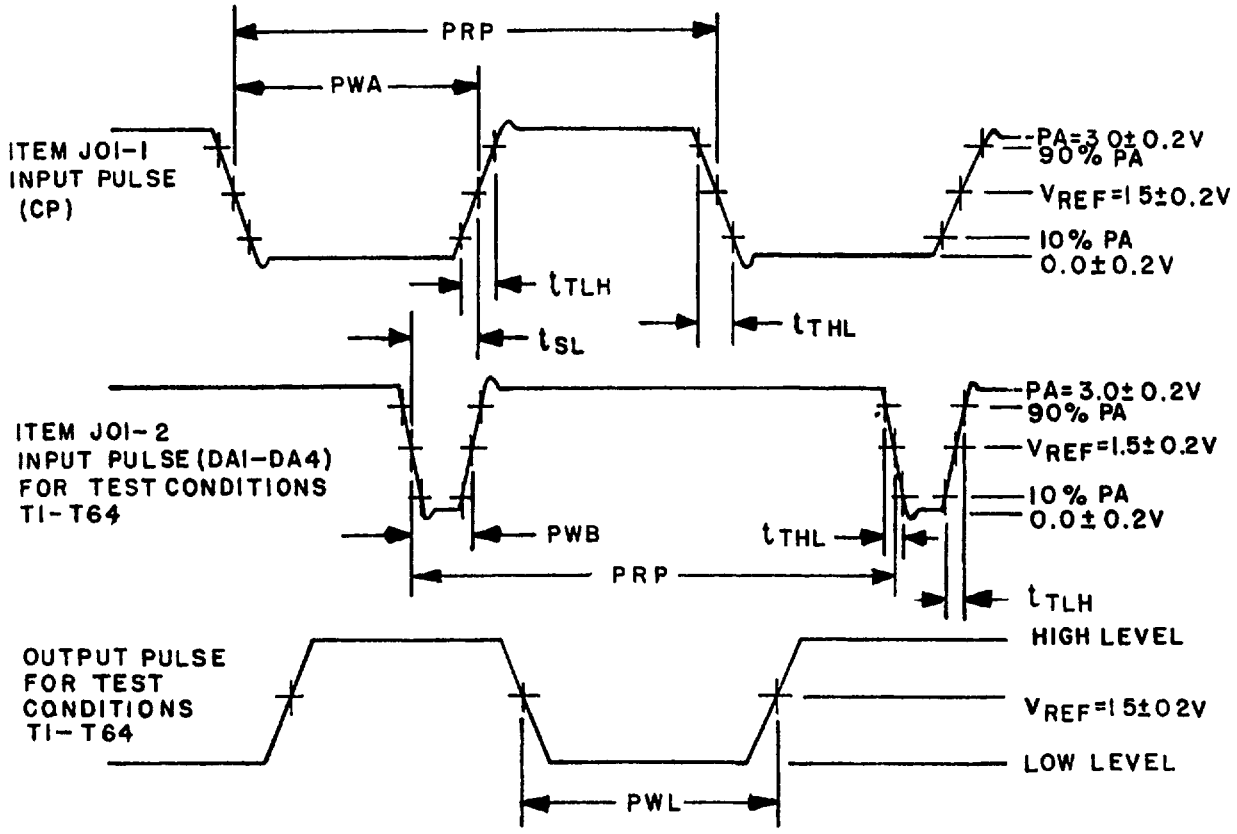
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NOTES:
1. Item J01-1 input pulse parameters (double pulse mode):
   PWB = 250 ±25 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
   t_D = 125 ±15 ns.
3. All voltages are referenced to ground.

FIGURE 22. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XVII and XVIII.

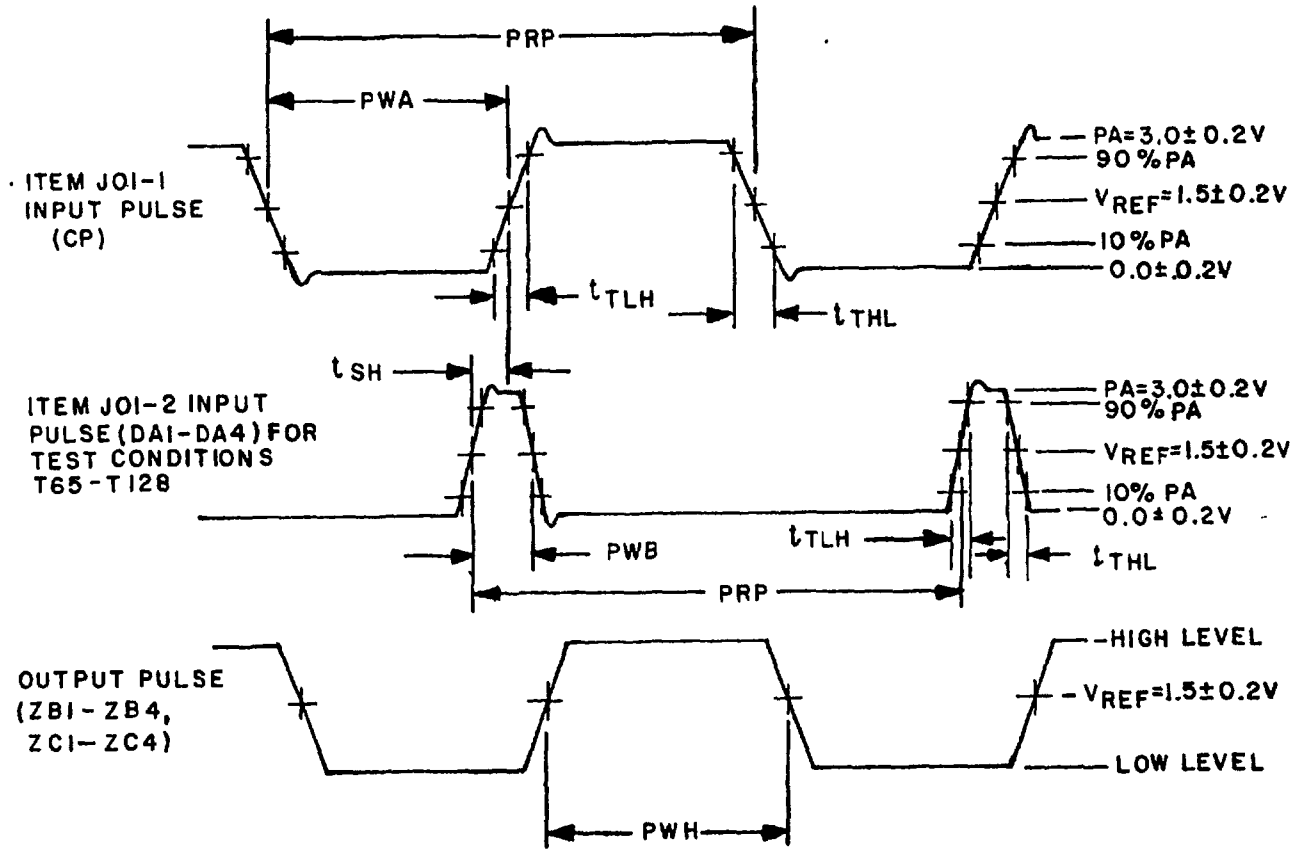
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:38Z
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NOTE: Input pulse parameters:
PRP = 500 ±50 ns.
PWA = 300 ±30 ns.
PWB = 30 ±3 ns.
t_{SL} = 12 -0, +2 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

FIGURE 23. Minimum setup and hold time (t_{SL}) test circuit waveforms for table XIX.

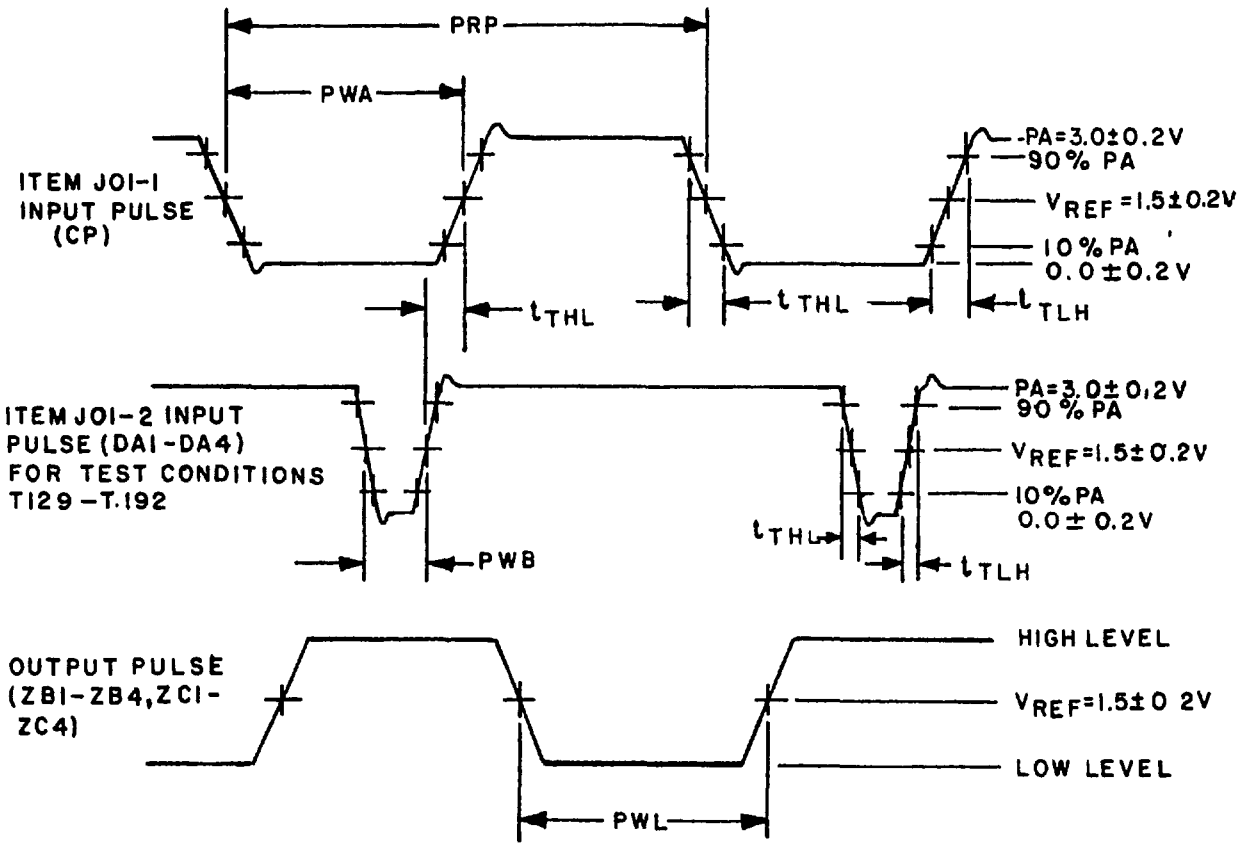
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:38Z
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NOTE: Input pulse parameters:
PRP = 500 ±50 ns.
PWA = 300 ±30 ns.
PWB = 30 ±2 ns.
t_{SH} = 20-0, +2 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

FIGURE 24. Minimum setup and hold time (t_{SH}) test circuit waveforms for table XIX.

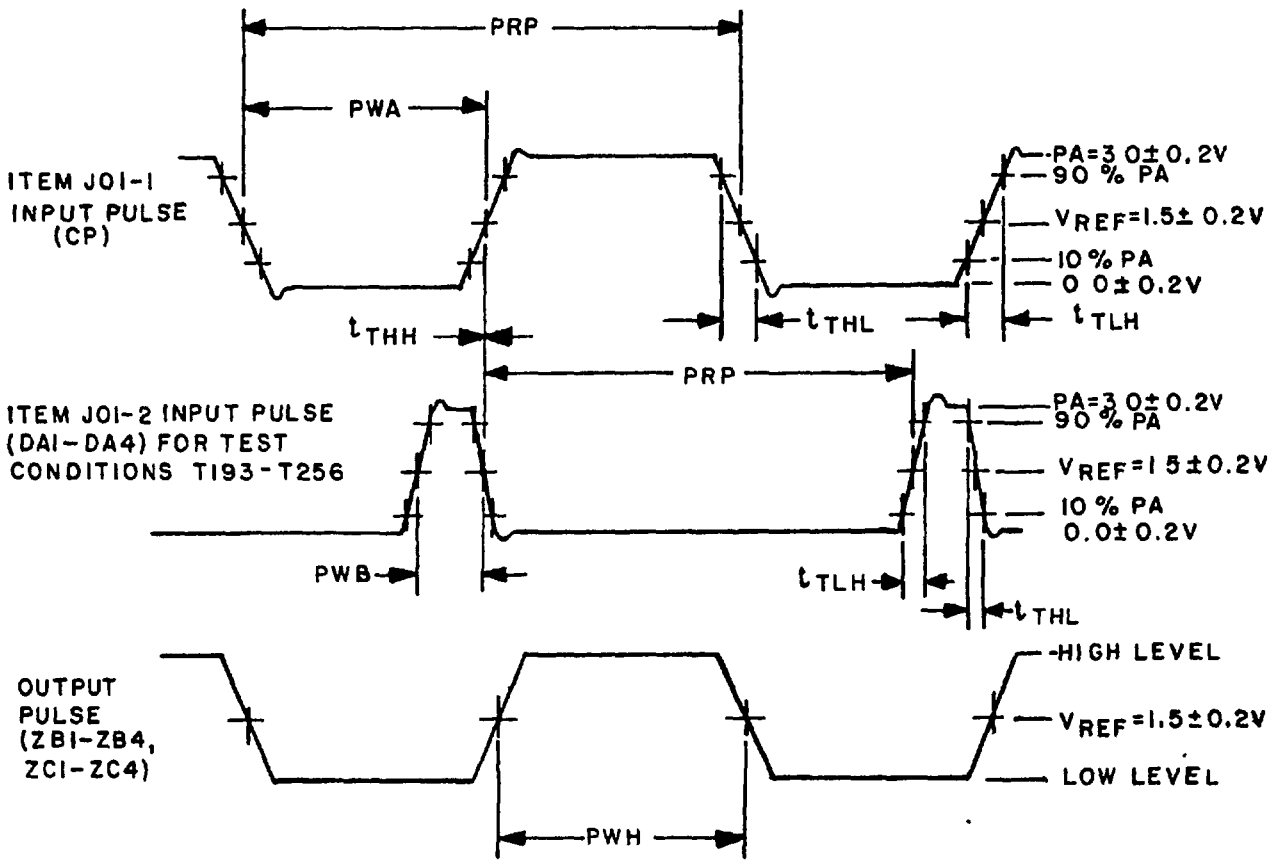
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:38Z
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NOTE: Input pulse parameters:
PRP = 500 ±50 ns.
PWA = 300 ±30 ns.
PWB = 30 ±2 ns.
t_{HL} = 8 -0, +2 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

FIGURE 25. Minimum setup and hold time (t_{THL}) test circuit waveforms for table XIX.

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NOTE: Input pulse parameters:
PRP = 500 ±50 ns.
PWA = 300 ±30 ns.
PWB = 30 ±3 ns.
t_{THH} = 0 -0, +2 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

FIGURE 26. Minimum setup and hold time (t_{HH}) test circuit waveforms for table XIX.

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TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U1	ZB1	Minimum clock pulse width (t_{CPL})	Open	14,40,17,33,37, 20,12,18,39,27	38	205	295	ns	27,28
U2	ZB2	"	"	"	34	"	"	"	"
U3	ZB3	"	"	"	30	"	"	"	"
U4	ZB4	"	"	"	4	"	"	"	"
U5	ZC1	"	"	"	19	"	"	"	"
U6	ZC2	"	"	"	15	"	"	"	"
U7	ZC3	"	"	"	11	"	"	"	"
U8	ZC4	"	"	"	24	"	"	"	"
U9	ZB1	"	14,40,17	33,37,20,12,18, 39,27	38	"	"	"	"
U10	ZB2	"	"	"	34	"	"	"	"
U11	ZB3	"	"	"	30	"	"	"	"
U12	ZB4	"	"	"	4	"	"	"	"
U13	ZC1	"	"	"	19	"	"	"	"
U14	ZC2	"	"	"	15	"	"	"	"
U15	ZC3	"	"	"	11	"	"	"	"
U16	ZC4	"	"	"	24	"	"	"	"
U17	ZB1	"	33,37,20	14,40,17,12,18, 39,27	38	"	"	"	"
U18	ZB2	"	"	"	34	"	"	"	"
U19	ZB3	"	"	"	30	"	"	"	"
U20	ZB4	"	"	"	4	"	"	"	"
U21	ZC1	"	"	"	19	"	"	"	"
U22	ZC2	"	"	"	15	"	"	"	"
U23	ZC3	"	"	"	11	"	"	"	"
U24	ZC4	"	"	"	24	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:38Z
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TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U25	ZB1	Minimum clock pulse width (t_{CPL})	14,40,17,33,37,20	27,12,18,39	38	205	295	ns	27,28
U26	ZB2	"	"	"	34	"	"	"	"
U27	ZB3	"	"	"	30	"	"	"	"
U28	ZB4	"	"	"	4	"	"	"	"
U29	ZC1	"	"	"	19	"	"	"	"
U30	ZC2	"	"	"	15	"	"	"	"
U31	ZC3	"	"	"	11	"	"	"	"
U32	ZC4	"	"	"	24	"	"	"	"
U33	ZB1	"	12,18,39	14,40,17,33,37, 20,27	38	"	"	"	"
U34	ZB2	"	"	"	34	"	"	"	"
U35	ZB3	"	"	"	30	"	"	"	"
U36	ZB4	"	"	"	4	"	"	"	"
U37	ZC1	"	"	"	19	"	"	"	"
U38	ZC2	"	"	"	15	"	"	"	"
U39	ZC3	"	"	"	11	"	"	"	"
U40	ZC4	"	"	"	24	"	"	"	"
U41	ZB1	"	12,18,39,14,40,17	27,33,37,20	38	"	"	"	"
U42	ZB2	"	"	"	34	"	"	"	"
U43	ZB3	"	"	"	30	"	"	"	"
U44	ZB4	"	"	"	4	"	"	"	"
U45	ZC1	"	"	"	19	"	"	"	"
U46	ZC2	"	"	"	15	"	"	"	"
U47	ZC3	"	"	"	11	"	"	"	"
U48	ZC4	"	"	"	24	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:38Z
Check the source to verify that this is the current version before use.

TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U49	ZB1	Minimum clock pulse width (t_{CPL})	33,37,20,12,18,39	27,14,40,17	38	205	295	ns	27,28
U50	ZB2	"	"	"	[illegible]	"	"	"	"
U51	ZB3	"	"	"	30	"	"	"	"
U52	ZB4	"	"	"	4	"	"	"	"
U53	ZC1	"	"	"	19	"	"	"	"
U54	ZC2	"	"	"	15	"	"	"	"
U55	ZC3	"	"	"	11	"	"	"	"
U56	ZC4	"	"	"	24	"	"	"	"
U57	ZB1	"	14,40,17,33,37, 20,12,18,39	27	38	"	"	"	"
U58	ZB2	"	"	"	34	"	"	"	"
U59	ZB3	"	"	"	30	"	"	"	"
U60	ZB4	"	"	"	4	"	"	"	"
U61	ZC1	"	"	"	19	"	"	"	"
U62	ZC2	"	"	"	15	"	"	"	"
U63	ZC3	"	"	"	11	"	"	"	"
U64	ZC4	"	"	"	24	"	"	"	"
U65	ZB1	Minimum clock pulse width (t_{CPH})	27	14,40,17,33,37, 20,12,18,39	38	"	"	"	"
U66	ZB2	"	"	"	34	"	"	"	"
U67	ZB3	"	"	"	30	"	"	"	"
U68	ZB4	"	"	"	4	"	"	"	"
U69	ZC1	"	"	"	19	"	"	"	"
U70	ZC2	"	"	"	15	"	"	"	"
U71	ZC3	"	"	"	11	"	"	"	"
U72	ZC4	"	"	"	24	"	"	"	"

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TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U73	ZB1	Minimum clock pulse width (t_{CPH})	27,14,40,17	33,37,20,12,18,39	38	205	295	ns	27,28
U74	ZB2	"	"	"	34	"	"	"	"
U75	ZB3	"	"	"	30	"	"	"	"
U76	ZB4	"	"	"	4	"	"	"	"
U77	ZC1	"	"	"	19	"	"	"	"
U78	ZC2	"	"	"	15	"	"	"	"
U79	ZC3	"	"	"	11	"	"	"	"
U80	ZC4	"	"	"	24	"	"	"	"
U81	ZB1	"	33,37,20,27	14,40,17,12,18,39	38	"	"	"	"
U82	ZB2	"	"	"	34	"	"	"	"
U83	ZB3	"	"	"	30	"	"	"	"
U84	ZB4	"	"	"	4	"	"	"	"
U85	ZC1	"	"	"	19	"	"	"	"
U86	ZC2	"	"	"	15	"	"	"	"
U87	ZC3	"	"	"	11	"	"	"	"
U88	ZC4	"	"	"	24	"	"	"	"
U89	ZB1	"	14,40,17,33,37, 20,27	12,18,39	38	"	"	"	"
U90	ZB2	"	"	"	34	"	"	"	"
U91	ZB3	"	"	"	30	"	"	"	"
U92	ZB4	"	"	"	4	"	"	"	"
U93	ZC1	"	"	"	19	"	"	"	"
U94	ZC2	"	"	"	15	"	"	"	"
U95	ZC3	"	"	"	11	"	"	"	"
U96	ZC4	"	"	"	24	"	"	"	"

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TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test - Continued.

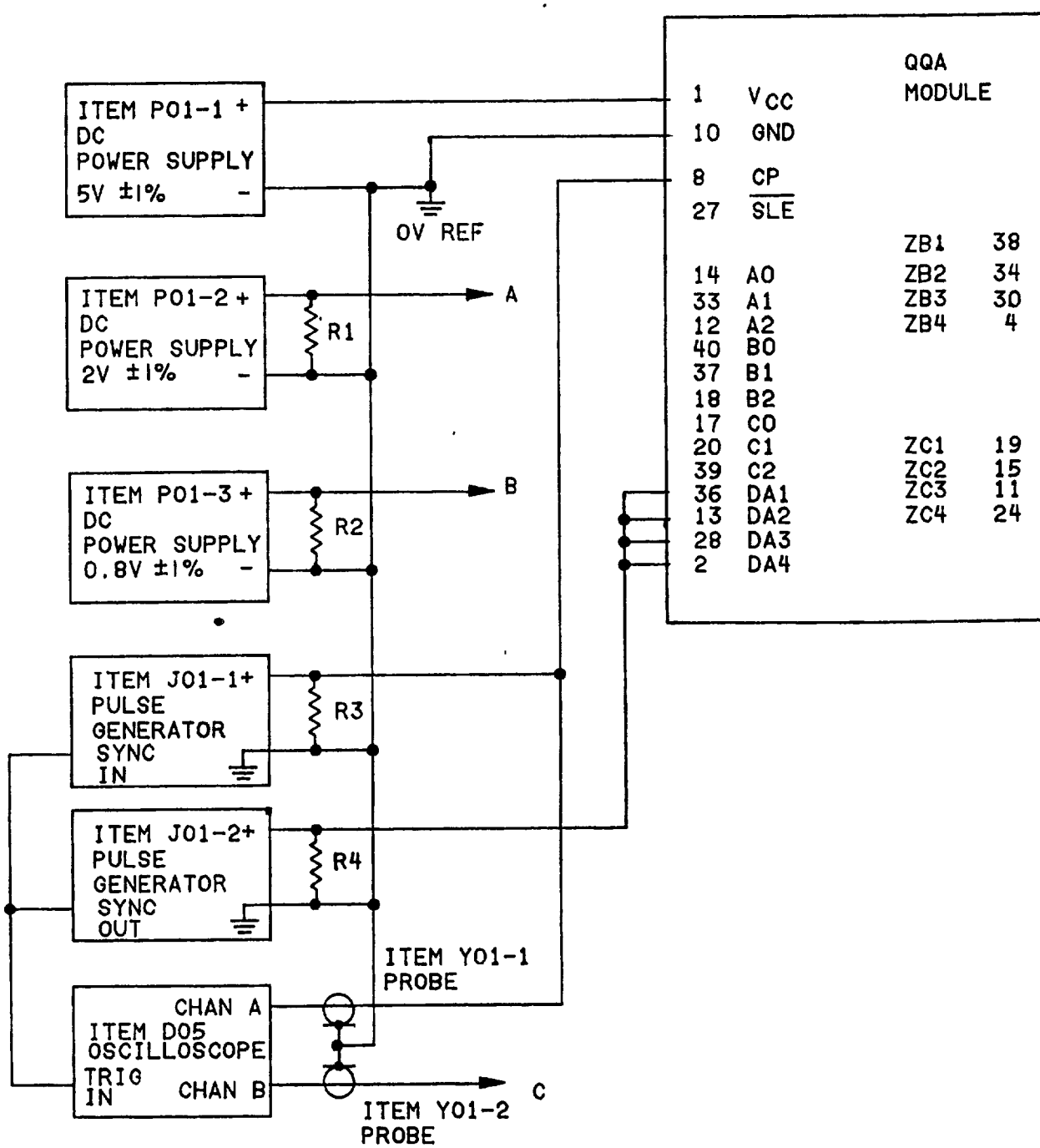
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U97	ZB1	Minimum clock pulse width (t_{CPH})	12,18,39,27	14,40,17,33,37,20	38	205	295	ns	27,28
U98	ZB2	"	"	"	34	"	"	"	"
U99	ZB3	"	"	"	30	"	"	"	"
U100	ZB4	"	"	"	4	"	"	"	"
U101	ZC1	"	"	"	19	"	"	"	"
U102	ZC2	"	"	"	15	"	"	"	"
U103	ZC3	"	"	"	11	"	"	"	"
U104	ZC4	"	"	"	24	"	"	"	"
U105	ZB1	"	12,18,39,14,40,17,27	33,37,20	38	"	"	"	"
U106	ZB2	"	"	"	34	"	"	"	"
U107	ZB3	"	"	"	30	"	"	"	"
U108	ZB4	"	"	"	4	"	"	"	"
U109	ZC1	"	"	"	19	"	"	"	"
U110	ZC2	"	"	"	15	"	"	"	"
U111	ZC3	"	"	"	11	"	"	"	"
U112	ZC4	"	"	"	24	"	"	"	"
U113	ZB1	"	33,37,20,12,18,39,27	14,40,17	38	"	"	"	"
U114	ZB2	"	"	"	34	"	"	"	"
U115	ZB3	"	"	"	30	"	"	"	"
U116	ZB4	"	"	"	4	"	"	"	"
U117	ZC1	"	"	"	19	"	"	"	"
U118	ZC2	"	"	"	15	"	"	"	"
U119	ZC3	"	"	"	11	"	"	"	"
U120	ZC4	"	"	"	24	"	"	"	"

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TABLE XX. Minimum clock pulse width (t_{CPL} and t_{CPH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U121	ZB1	Minimum clock pulse width (t_{CPH})	14,40,17,33,37, 20,12,18,39,27	Open	38	205	295	ns	27,28
U122	ZB2	"	"	"	34	"	"	"	"
U123	ZB3	"	"	"	30	"	"	"	"
U124	ZB4	"	"	"	4	"	"	"	"
U125	ZC1	"	"	"	19	"	"	"	"
U126	ZC2	"	"	"	15	"	"	"	"
U127	ZC3	"	"	"	11	"	"	"	"
U128	ZC4	"	"	"	24	"	"	"	"

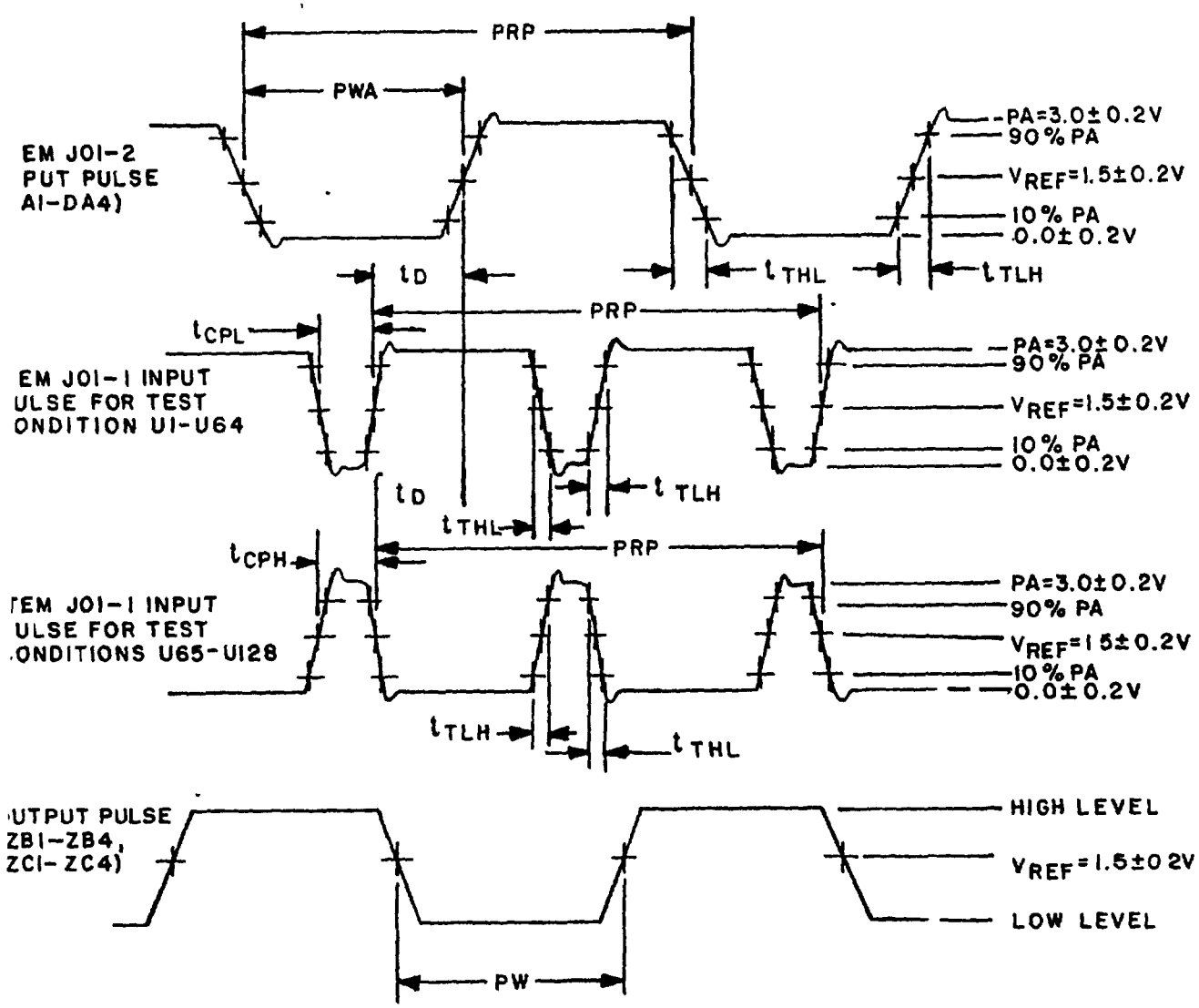
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NOTE: R1-R4 = 51Ω ±5%, 1/4 W.

FIGURE 27. Minimum clock pulse width (t_{CPL} and t_{CPH}) test circuit for table XX.

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NOTE: Input pulse parameters:
PRP = 500 ±50 ns.
PWA = 250 ±25 ns.
t_{CPL} = 13 -0, +2 ns.
t_{CPH} = 23 -0, +2 ns.
t_D = 100 ±10 ns.

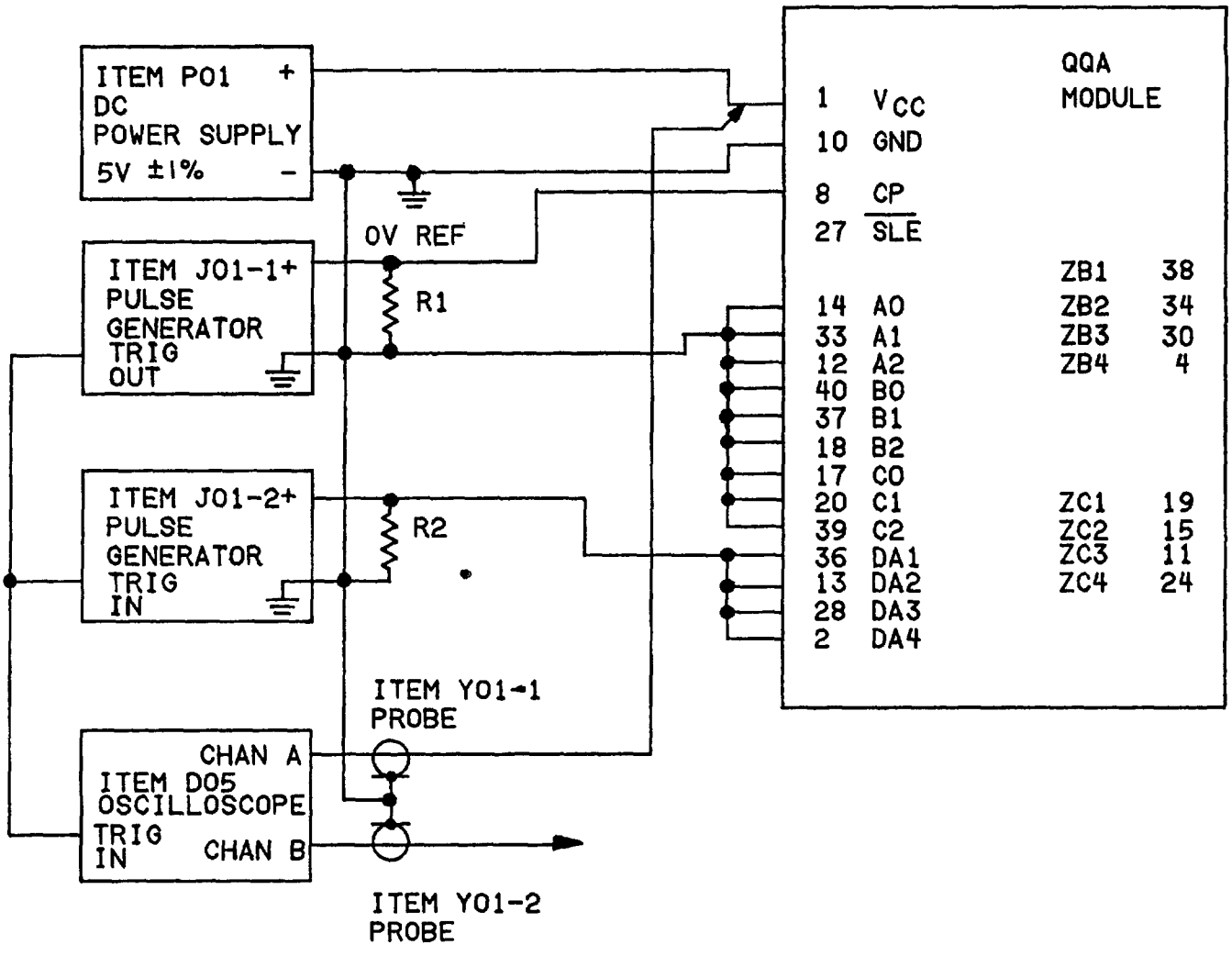
FIGURE 28. Minimum clock pulse width (t_{CPL} and t_{CPH}) test circuit waveforms for table XX.

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TABLE XXI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	29,30

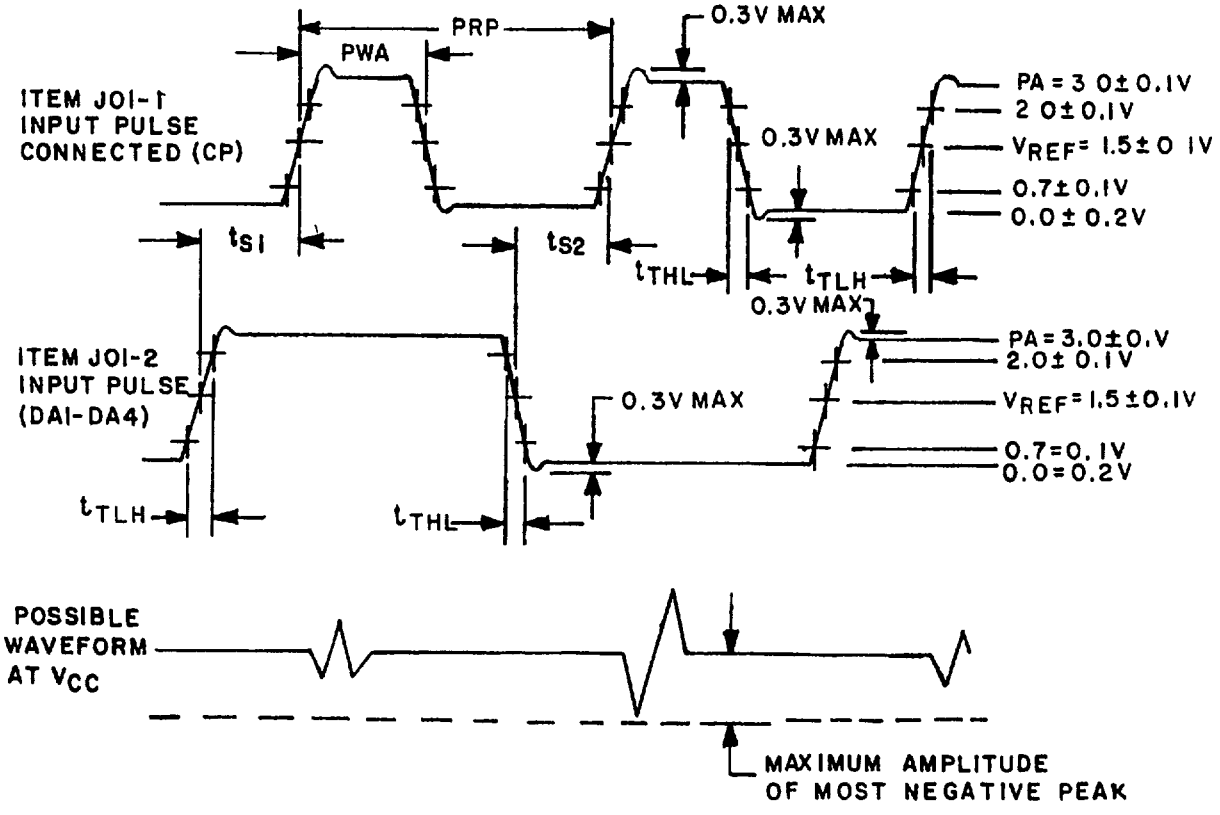
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 29. Power supply transient amplitude test circuit for table XXI.

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NOTE: Input pulse parameters:
PWA = 250 ±25 ns.
PRP = 500 ±50 ns.
$t_{S1} = t_{S2}$ = 50 ±5 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

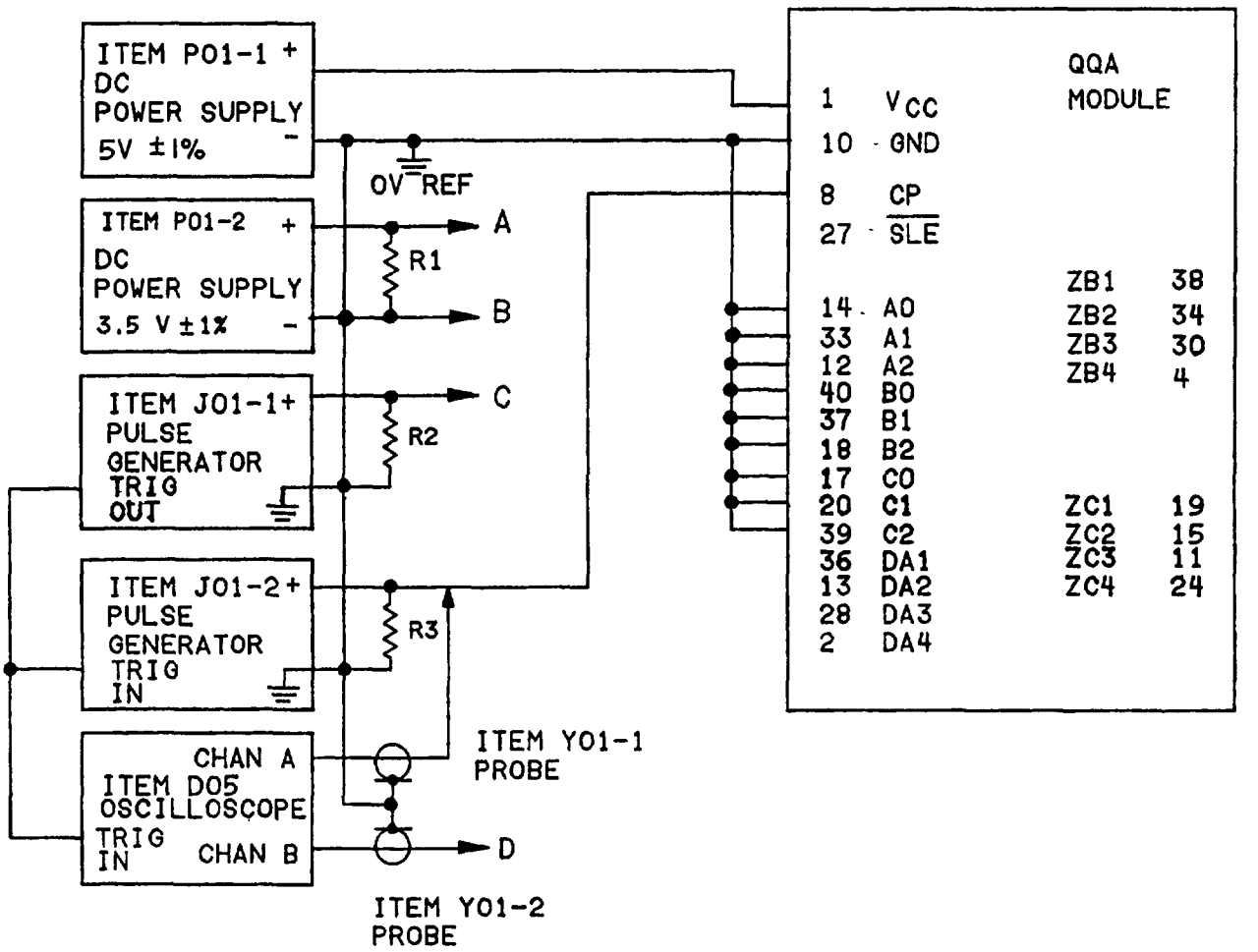
FIGURE 30. Power supply transient amplitude test circuit waveforms for table XXI.

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TABLE XXII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	ZB1	Output crosstalk, high level	36	Open	13,28,2	38	0	400	mV	31,32
W2	ZB2	"	13	"	36,28,2	34	"	"	"	"
W3	ZB3	"	28	"	36,13,2	30	"	"	"	"
W4	ZB4	"	2	"	36,28,13	4	"	"	"	"
W5	ZC1	"	36	"	13,28,2	19	"	"	"	"
W6	ZC2	"	13	"	36,28,2	15	"	"	"	"
W7	ZC3	"	28	"	36,13,2	11	"	"	"	"
W8	ZC4	"	2	"	36,28,13	24	"	"	"	"
X1	ZB1	Output crosstalk, low level	Open	36	13,28,2	38	"	"	"	"
X2	ZB2	"	"	13	36,28,2	34	"	"	"	"
X3	ZB3	"	"	28	36,13,2	30	"	"	"	"
X4	ZB4	"	"	2	36,28,13	4	"	"	"	"
X5	ZC1	"	"	36	13,28,2	19	"	"	"	"
X6	ZC2	"	"	13	36,28,2	15	"	"	"	"
X7	ZC3	"	"	28	36,13,2	11	"	"	"	"
X8	ZC4	"	"	2	36,28,13	24	"	"	"	"

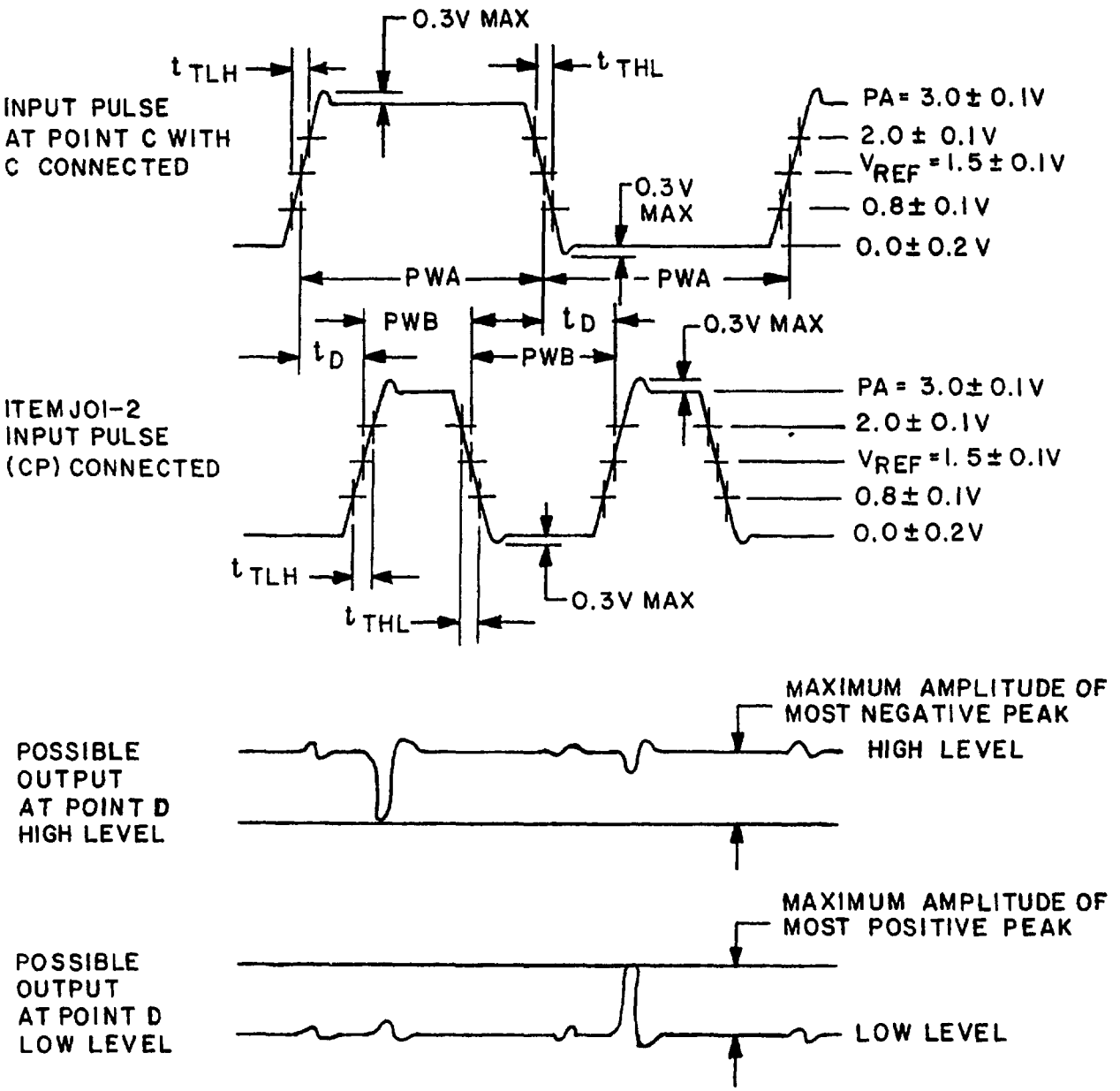
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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1/4 W.
2. Load all outputs with load as shown on figure 15.

FIGURE 31. Output crosstalk (low-level and high-level) test circuit for table XXII.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   t_D = 100 ±10 ns.
2. Before making the measurements specified in table XIX, verify that the ouput pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 v.
3. The lead length of electrical conductors between resistors R2 and R3 and input pins, between probes Y01-1 and Y01-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).

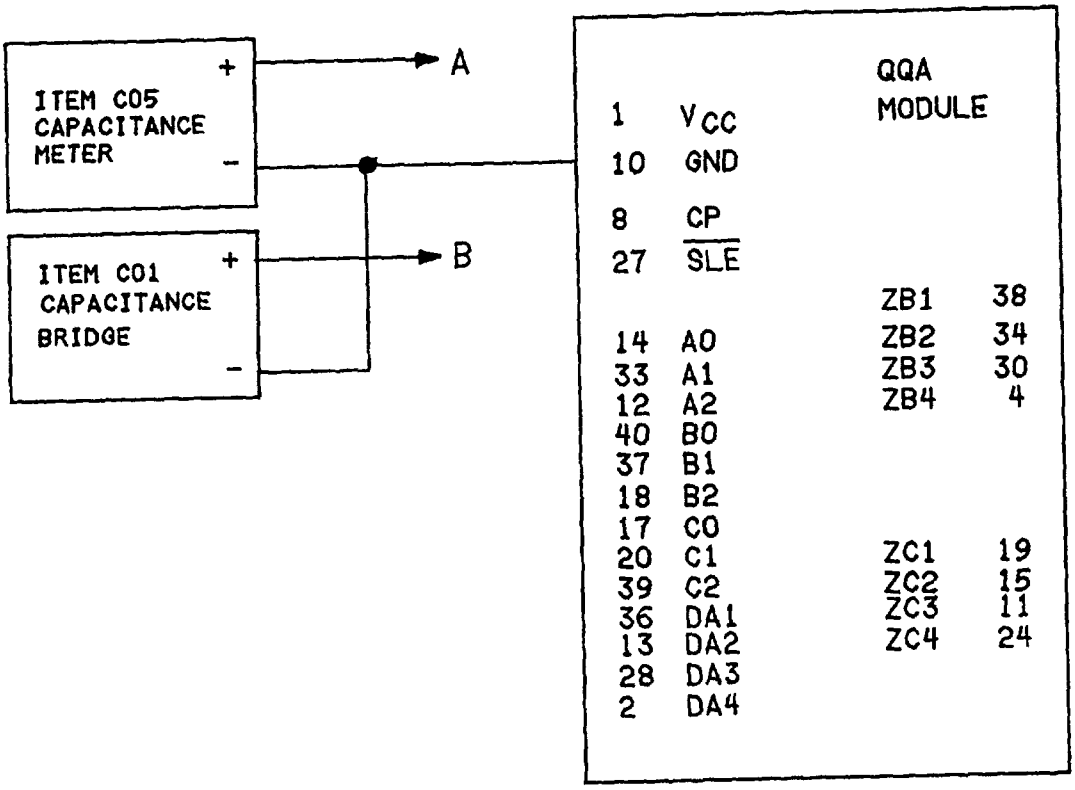
FIGURE 32. Output crosstalk (low-level and high-level) test circuit waveforms for table XXII.

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TABLE XXIII. Power supply filter capacitance and input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	Open	1	1.98	2.42	µF	33
Z1	CP Input capacitance	8	Open	8	60	pF	"
Z2	$\overline{SLE}$ "	27	"	"	"	"	"
Z3	A0 "	14	"	"	"	"	"
Z4	A1 "	33	"	"	"	"	"
Z5	A2 "	12	"	"	"	"	"
Z6	B0 "	40	"	"	"	"	"
Z7	B1 "	37	"	"	"	"	"
Z8	B2 "	18	"	"	"	"	"
Z9	C0 "	17	"	"	"	"	"
Z10	C1 "	20	"	"	"	"	"
Z11	C2 "	39	"	"	"	"	"
Z12	DA1 "	36	"	2	15	"	"
Z13	DA2 "	13	"	"	"	"	"
Z14	DA3 "	28	"	"	"	"	"
Z15	DA4 "	2	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 33. Power supply filter capacitance and input capacitance test circuit for table XXIII.

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